

Four Wheel Drive (4WD) Systems — 2-Speed Torque-On-Demand

 [Ford Ecat Electronic Parts Catalog](#)

Special Tool(s)	
 ST2574-A	Flex Probe Kit 300-NUD105-R025DE or equivalent
 ST3093-A	Fluke 77-IV Digital Multimeter FLU77-4 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool

Principles of Operation

The 2-speed torque-on-demand system is an automatic electronic-shift system that allows the operator to choose between 2-Wheel Drive (2WD) and 3 different Four-Wheel Drive (4WD) modes. The operator can switch between 2WD (2H) mode, 4X4 AUTO (4A) mode or 4X4 (4H) mode at any speed or at a stop. To engage or disengage 4X4 LOW (4L) mode, the vehicle speed must be less than 5 km/h (3 mph) and the transmission must be in NEUTRAL.

Engagement and disengagement of the front wheels is done by the IWE s. The IWE s are normally engaged (no vacuum). Vacuum is supplied to the IWE s by actuation of the IWE solenoid commanded by the IWE . When actuated, the IWE solenoid allows vacuum to pass through to each IWE . The IWE s disengage and allow the front wheel hubs to be turned freely from the front axle

NOTE: When shifting to 2H mode in order for the IWEs to completely disengage the vehicle steering wheel must be less than 90 degrees and 5 seconds have elapsed since the shift to 2H has been initiated.

Wheel slip is sensed by monitoring the average front and rear wheel speeds. Based on steering wheel angle, Accelerator Pedal Position (APP) and wheel slip, a Pulse-Width Modulation (PWM) signal is transmitted to the transfer case as torque is required.

The Transfer Case Control Module (TCCM) will provide the ABS with current clutch duty cycle and whether or not ABS may command the clutch duty cycle.

The brake subsystem sends the following information signals to the 4WD system over the High Speed Controller Area Network (HS-CAN) :

- Occurrences of brake events and failures. Refer to Section 206-09.

- Steering wheel angle status. Refer to Section 206-09.
- Wheel speeds. Refer to Section 206-09.

A concern with any of the above listed brake system signals ~~HS-CAN~~ will affect operation of the ~~4WD~~ system.

The ~~APP~~ output signal is provided to the ~~TCCM~~ from the PCM. This signal is used by the ~~TCCM~~ in controlling the transfer case clutch.

The shift motor sense plate, an integral part of the shift motor assembly, informs the ~~TCCM~~ of the transfer case position.

The electric shift motor is mounted externally to the transfer case. It drives a rotary cam which moves the range fork within the transfer case between the HIGH range (2H, 4A and 4H) and LOW range (4L) positions. **The shift motor only moves during a shift to/from 4L on the 2 speed torque on demand system; it does not move during any other mode shift.**

The ~~TCCM~~ sends a duty cycle command to the clutch coil which activates the clutch within the transfer case.

The ~~TCCM~~ varies the torque sent to the front driveline by controlling the transfer case clutch. At rest and under cruising conditions, the ~~TCCM~~ activates the transfer case clutch a minimum duty cycle (percentage of time the clutch is turned on), which allows for the slight difference between the front and rear driveshafts which normally occurs when negotiating a corner on dry pavement. Under any of the following conditions, the ~~TCCM~~ will increase the duty cycle in order to prevent or control slip:

- Slip is detected
- Heavy acceleration (throttle position)
- Straight-ahead steering wheel angle

Feature inputs:

- ~~MSS~~
- Steering wheel angle (from the ABS module)
- ~~TR~~ position (from the PCM)
- Transmission output torque (from the PCM)
- Wheel speed signal (transmitted from the ABS module)
- ~~APP~~ (from the PCM)
- Transfer case shift motor contact plate position inputs A, B, C, D

Feature outputs:

- ~~PWM~~ signal to the transfer case clutch
- ~~4WD~~ message center indicators
- Transfer case shift motor outputs

Shifts between 4X4 AUTO (4A) and 4X4 (4H) modes can be made at any speed. Listed below are the inputs and outputs needed by the to execute a change between any of these modes.

Feature inputs:

- Wheel speed signal
- ~~APP~~ (from the PCM)
- ~~MSS~~

Feature outputs:

- Transfer case shift motor outputs
- ~~PWM~~ signal to the transfer case clutch
- ~~4WD~~ message center indicators

When shifting into or out of 4X4 LOW (4L) mode, ~~TCCM~~ the requires that the vehicle speed is less than 5 km/h (3 mph) and the transmission is in NEUTRAL.

Feature inputs:

- APP (from the PCM)
- MSS
- Transfer case shift motor encoder contact plate position inputs A, B, C, D
- Wheel speed signal (transmitted from ABS module)

Feature outputs:

- Transfer case shift motor outputs
- 4WD message center indicators
- PWM signal to the transfer case clutch

Inspection and Verification

1. Verify the customer concern.
2. Visually inspect for obvious signs of mechanical or electrical damage.

Mechanical	Electrical
• Halfshafts • IWEs • Driveshaft and U-joints • Vacuum leaks • Fluid leaks • Matching tire sizes • Transfer case	• BJB fuses: ▪ 20 (20A) ▪ 29 (10A) ▪ 54 (5A) ▪ 68 (25A) • BCM fuse 34 (10A) (vehicles with ELD only) • Wiring, terminals or connectors • MSS • Synchronization clutch • IWE solenoid • Transfer case shift motor

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.
4. **NOTE:** *Make sure to use the latest scan tool software release.*

If the cause is not visually evident, connect the scan tool to the Data Link Connector (DLC) .

5. **NOTE:** *The Vehicle Communication Module (VCM) LED prove out confirms power and ground from the DLC are provided to the VCM .*

If the scan tool does not communicate with the VCM :

- check the VCM connection to the vehicle.
- check the scan tool connection to the VCM .
- refer to Section 418-00, No Power To The Scan Tool, to diagnose no power to the scan tool.

6. If the scan tool does not communicate with the vehicle:
 - verify the ignition is in the ON position.
 - verify the scan tool operation with a known good vehicle.

- refer to Section 418-00 to diagnose no response from the PCM.

7. Carry out the network test.

- If the scan tool responds with no communication for one or more modules, refer to Section 418-00.
- If the network test passes, retrieve and record Continuous Memory Diagnostic Trouble Codes (CMDTCs) .

8. Carry out the self-test diagnostics for the ~~TCCM~~ .

9. If the DTCs retrieved are related to the concern, go to the ~~TCCM~~ DTC Chart in this section. For all other DTCs, refer to Section 419-10.

10. If no DTCs related to the concern are retrieved, GO to [Symptom Chart — Four-Wheel Drive System](#) or GO to [Symptom Chart — NVH](#).

DTC Charts

Transfer Case Control Module (TCCM) DTC Chart

NOTE: This module utilizes a 5-character DTC followed by a 2-character failure type code. The failure type code provides information about specific fault conditions such as opens or shorts to ground. Continuous Memory Diagnostic Trouble Codes (CMDTCs) have an additional 2-character DTC status code suffix to assist in determining DTC history.

DTC	Description	Action
C1979:11	IWE Solenoid Circuit Failure: Circuit Short To Ground	GO to Pinpoint Test N.
C1979:12	IWE Solenoid Circuit Failure: Circuit Short To Battery	GO to Pinpoint Test N.
C1979:13	IWE Solenoid Circuit Failure: Circuit Open	GO to Pinpoint Test N.
P0562:00	System Voltage Low: No Sub Type Information	GO to Pinpoint Test M.
P0563:00	System Voltage High: No Sub Type Information	GO to Pinpoint Test M.
P1812:00	4-Wheel Drive Mode Select Circuit Failure: No Sub Type Information	GO to Pinpoint Test X.
P1815:00	4-Wheel Drive Mode Select Short To Ground: No Sub Type Information	GO to Pinpoint Test X.
P1825:00	4-Wheel Drive Clutch Relay Open Circuit: No Sub Type Information	GO to Pinpoint Test Q.
P1826:00	4-Wheel Drive Low Clutch Relay Circuit To Battery: No Sub Type Information	GO to Pinpoint Test Q.
P1827:00	4-Wheel Drive Low Clutch Relay Circuit To Ground: No Sub Type Information	GO to Pinpoint Test Q.
P1839:00	Transfer Case Shift Motor Open Circuit: No Sub Type Information	GO to Pinpoint Test P.
P1840:00	Transfer Case Shift Motor Short Circuit To Battery: No Sub Type Information	GO to Pinpoint Test P.
P1841:00	Transfer Case Shift Motor Short Circuit To Ground: No Sub Type Information	GO to Pinpoint Test P.
P1849:00	Transfer Case Contact Plate 'A' Short Circuit To Ground: No Sub Type Information	GO to Pinpoint Test P.

DTC	Description	Action
P184B:00	Transfer Case Unable To Transition Between 4H and 4L: No Sub Type Information	GO to Pinpoint Test P.
P1853:00	Transfer Case Contact Plate 'B' Short Circuit To Ground: No Sub Type Information	GO to Pinpoint Test P.
P1857:00	Transfer Case Contact Plate 'C' Short Circuit To Ground: No Sub Type Information	GO to Pinpoint Test P.
P185A:00	Differential Lock-Up Input Switch Circuit High: No Sub Type Information	Go to Pinpoint Test Y in Electronic Locking Differential (ELD) System, Diagnosis and Testing, Section 308-07A.
P185B:00	Differential Lock-Up Indicator Circuit Low: No Sub Type Information	Go to Pinpoint Test Z in Electronic Locking Differential (ELD) System, Diagnosis and Testing, Section 308-07A.
P185C:00	Differential Lock-Up Indicator Circuit High: No Sub Type Information	Go to Pinpoint Test AA in Electronic Locking Differential (ELD) System, Diagnosis and Testing, Section 308-07A.
P185D:00	Differential Lock-Up Control Circuit Performance: No Sub Type Information	Go to Pinpoint Test AB in Electronic Locking Differential (ELD) System, Diagnosis and Testing, Section 308-07A.
P1861:00	Transfer Case Contact Plate 'D' Short Circuit To Ground: No Sub Type Information	GO to Pinpoint Test P.
P1867:00	Transfer Case Contact Plate General Circuit Failure: No Sub Type Information	GO to Pinpoint Test P.
P187C:00	Differential Lock-Up Control Circuit/Open: No Sub Type Information	Go to Pinpoint Test AC in Electronic Locking Differential (ELD) System, Diagnosis and Testing, Section 308-07A.
P187D:00	Differential Lock-Up Control Circuit Low: No Sub Type Information	Go to Pinpoint Test AC in Electronic Locking Differential (ELD) System, Diagnosis and Testing, Section 308-07A.
P187E:00	Differential Lock-Up Control Circuit High: No Sub Type Information	Go to Pinpoint Test AD in Electronic Locking Differential (ELD) System, Diagnosis and Testing, Section 308-07A.
P1886:00	4X4 Initialization Failure: No Sub Type Information	CLEAR the DTC. REPEAT the self-test. If the DTC still reports, CONFIRM the TCCM part number is correct for the vehicle.
P1891:00	Transfer Case Contact Plate Ground Return Open Circuit: No Sub Type Information	GO to Pinpoint Test P.
U0100:00	Lost Communication With ECM/PCM "A": No Sub Type Information	Go to Pinpoint Test AB in Electronic Locking Differential (ELD) System, Diagnosis and Testing, Section 308-07A.
U0121:00	Lost Communication With Anti-Lock Brake System (ABS) Control Module: No Sub Type Information	Go to Pinpoint Test AF in Electronic Locking Differential (ELD) System, Diagnosis and Testing, Section 308-07A.
U0418:00	Invalid Data Received From Brake System Control Module: No Sub Type Information	REFER to Section 206-09 to diagnose the wheel speed sensor concern.
U3000:04	Control Module: System Internal Failure	CLEAR the DTC. REPEAT the self-test. If DTC U3000:04 is retrieved, INSTALL a new TCCM . REFER to Transfer Case Control Module (TCCM) in this section.
U3000:41	Control Module: General Checksum Failure	CLEAR the DTC. REPEAT the self-test. If DTC U3000:41 is retrieved, INSTALL a new TCCM . REFER to Transfer Case Control Module (TCCM) in this section.
U3000:46	Control Module: Calibration / Parameter Memory Failure	CLEAR the DTC. REPEAT the self-test. If DTC U3000:46 is retrieved, INSTALL a new TCCM . REFER to Transfer Case Control Module (TCCM) in this section.
U3000:49	Control Module: Internal Electronic Failure	CLEAR the DTC. REPEAT the self-test. If DTC U3000:49 is retrieved, INSTALL a new TCCM . REFER to Transfer Case

DTC	Description	Action
		Control Module (TCCM) in this section.
U300D:11	Ignition Input On/Start: Circuit Short To Ground	GO to Pinpoint Test L.

Symptom Chart — Four-Wheel Drive System

Symptom Chart —		
Condition	Possible Causes	Action
No communication with the TCCM	- Scan tool - DLC - TCCM - Wiring, terminals or connectors	REFER to Section 418-00.
The 4X4 mode indicators do not operate correctly/do not operate	- Instrument Panel Cluster (IPC) 4WD system fault - HS-CAN - TCCM	GO to Pinpoint Test V.
The vehicle does not shift between 2H (2WD) and 4X4 modes correctly	- Wiring, terminals or connectors - MSS - Transfer case - TCCM	GO to Pinpoint Test X.
The vehicle does not shift between 4A (4X4 AUTO) and 4H (4X4) modes correctly	- MSS - IWEs - IWE solenoid - Transfer case - Vacuum leaks - Front axle assembly - TCCM - Wiring, terminals or connectors	GO to Pinpoint Test O.
Vehicle has no torque or inadequate torque at front	Transfer case and related components	CARRY OUT the 4WD System Functional Test. GO to Pinpoint

wheels in 4A (4X4 AUTO) mode	• IWEs and related components • Wheel/tire assemblies	Test U.
• The vehicle does not shift between 4H (4X4) and 4L (4X4 LOW) modes correctly	• Wiring, terminals or connectors • Transfer case • TR • ABS wheel speed inputs • IWEs • TCCM • Transmission drag torque	• GO to Pinpoint Test P.
• 4WD does not engage at speed correctly	• Transfer case synchronization clutch • Transfer case • TCCM • Front axle assembly • Wiring, terminals or connectors	• GO to Pinpoint Test Q.
• The front axle does not engage/disengage correctly	• MSS • IWEs • IWE solenoid • Transfer case • Vacuum leaks • Front axle assembly • TCCM • Wiring, terminals or connectors • Fuses	• GO to Pinpoint Test N.
• The 4WD system jumps out of gear	• IWE vent line(s) • MSS	• GO to Pinpoint Test R.

	• Transfer case	
• Unable to duplicate customer concern	• 4WD system and/or related components	• CARRY OUT the 4WD System Functional Test. GO to Pinpoint Test U.
• Unable to achieve flat tow (neutral tow) mode	• Wiring, terminals or connectors • HS-CAN • IPC • TCCM • Door ajar switch • Ignition switch	• Go to Pinpoint Test AH in Electronic Locking Differential (ELD) System, Diagnosis and Testing, Section 308-07A.

Symptom Chart — NVH

NOTE: NVH symptoms should be identified using the diagnostic tools that are available. For a list of these tools, an explanation of their uses and a glossary of common terms, refer to Section 100-04. Since it is possible any one of multiple systems may be the cause of a symptom, it may be necessary to use a process of elimination type of diagnostic approach to pinpoint the responsible system. If this is not the causal system for the symptom, refer back to Section 100-04 for the next likely system and continue diagnosis.

Symptom Chart — NVH

Condition	Possible Causes	Action
• Grinding noise during 4WD system engagement, especially at high speeds	• Front halfshaft speeds not turning at the same speed • Synchronization clutch	• GO to Pinpoint Test N.
• Vehicle Pulsates or Shudders in a Straight Line (Straightline Driveline Wind-up)	• Unequal tire inflation pressures • Unmatched tire and wheel size • Unequal amounts of tire wear • Unmatched front and rear axle ratios	• GO to Pinpoint Test S.

• The front axle makes noise in 2H mode only and generally under heavy throttle	• Vacuum leaks • Vacuum check valve • Vacuum reservoir • IWE solenoid • IWEs	• GO to Pinpoint Test T.
• The transfer case makes noise	• Tire inflation pressure • Tire and wheel size • Tire tread wear • Internal components • Fluid level	• MAKE SURE that all tires and wheels are the same size and that the inflation pressures are correct. • CHECK tire tread wear to see if there is more than 1.6 mm (0.062 in) or 2/32 inch difference in tread wear between front and rear. INTERCHANGE one front wheel and one rear wheel. ROAD TEST again. • OPERATE the vehicle in all transmission gears. If there is noise in the transmission in NEUTRAL, or in some gears and not in others, REMOVE and REPAIR the transmission. REFER to Section 307-01. If there is noise in all gears, DISASSEMBLE the transfer case. REFER to Section 308-07B. CHECK the planetary gears, the bearings, the upper and lower drive sprockets and drive chain for damage. INSTALL new parts as necessary. • FILL with automatic transmission fluid. REFER to Section 308-07B.
• Transfer case whine — noise at all ranges	• Incorrect fluid level or fluid quality • Worn oil pump • Under-inflated or oversized tires	• CHECK that the transfer case is filled to the correct level and with the specified fluid. REFER to Section 308-07B. • DISASSEMBLE the transfer case. CHECK the oil pump for wear or damage. REPAIR as necessary. REFER to Section 308-07B. • CONFIRM that the tires and wheels are correct for the vehicle. CHECK that the tire inflation pressures are correct.
• Transfer case growl/rumble — noise at all ranges	• Damaged or worn bearings or planetary gear	• DISASSEMBLE the transfer case. CHECK the bearings or planetary gear for wear or damage. REPAIR as necessary. REFER to Section 308-07B.
• Transfer case scraping/grating — noise at all ranges	• Excessively stretched drive chain hitting the case	• DISASSEMBLE the transfer case. CHECK the drive chain for wear or damage. REPAIR as necessary. REFER to Section 308-07B.

Transfer case howl/hum — noise at all ranges or 4H only	Worn or damaged sun (input) gear, clutch pack (intermediate) gear or output shaft gear	DISASSEMBLE the transfer case. CHECK the gears for wear or damage. REPAIR as necessary. REFER to Section 308-07B.
Vehicle binds in a turn/resists turning/pulsates in a straightline in 4WD mode	- Unmatched tire sizes - Unequal amounts of tire wear - Unequal tire inflation pressures - Throttle position failure - Transfer case - TCCM	GO to Pinpoint Test W.

Pinpoint Tests

Pinpoint Test L: DTC U300D:11

Refer to Wiring Diagrams Cell [34](#) , Four-Wheel Drive Systems for schematic and connector information.

Normal Operation

The TCCM receives ignition switch position status from the hardwired run/start input circuit and the ignition switch position status message communicated by the BCM over the HS-CAN .

- DTC U300D:11 (Ignition Input On/Start: Circuit Short To Ground) — Fault sets when short to ground is sensed by TCCM C2371A pin 7. If the ignition switch is not in RUN for the self-test, this code will set.

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- Ignition switch
- HS-CAN

PINPOINT TEST L : U300D:11

L1 CHECK THE TCCM IGNITION POSITION (KEY_POS_4X4) AND IGNITION STATUS (KEY_STATUS_4X4) PIDS	
- Enter the following diagnostic mode on the scan tool: DataLogger — TCCM . - Monitor the KEY_POS_4X4 and KEY_STATUS_4X4 PIDs.	
Do the ignition PIDs match the ignition switch positions?	
Yes	The system is functioning correctly at this time. CLEAR the DTCs. TEST the system for normal operation.
No	If the KEY_STATUS_4X4 PID did not match, CHECK BJB fuse 54 (5A). If the fuse is OK, REPAIR circuit CBB54 (VT/OG). If the fuse is not OK, REFER to the Wiring Diagrams manual to

identify the possible causes of the circuit short. After the fault is repaired, CLEAR the DTCs. TEST the system for normal operation.
If the KEY_POS_4X4 PID did not match, REFER to Section 211-05 to diagnose the ignition switch. After the fault is repaired, CLEAR the DTCs. TEST the system for normal operation.

Pinpoint Test M: DTC P0562:00 or P0563:00

Refer to Wiring Diagrams Cell [34](#) , Four-Wheel Drive Systems for schematic and connector information.

Normal Operation

If the TCCM observes an overpower or underpower voltage condition, a DTC is set and all TCCM outputs are turned off (IWEs engaged).

- DTC P0562:00 (Battery Voltage Low: No Sub Type Information) — Fault sets when the TCCM detects battery voltage less than 9 volts for 5 seconds or more at TCCM C2371B pin 7.
- DTC P0563:00 (Battery Voltage High: No Sub Type Information) — Fault sets when the TCCM detects battery voltage greater than 16 volts for 5 seconds or more at TCCM C2371B pin 7.

This pinpoint test is intended to diagnose the following:

- Fuses
- Wiring, terminals or connectors
- TCCM

PINPOINT TEST M : DTC P0562:00 OR P0563:00

NOTE: *DTC P0562:00 or P0563:00 can be set if the vehicle has been recently jump started, the battery has been recently charged or the battery has been discharged. The battery may become discharged due to excessive load(s) on the charging system from aftermarket accessories or if the battery has been left unattended with the accessories on.*

NOTE: *Carry out a thorough inspection and verification before proceeding with the pinpoint test. Refer to Inspection and Verification in this section.*

M1 RETRIEVE ALL CMDTCS IN ALL MODULES

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self-Test — All CMDTCS .

Is P0562:00 or P0563:00 present in one or more modules AND P0563, P0620, P0625, P0626 or P065B present in the PCM?

Yes	REFER to Section 414-00 for diagnosis of the battery and charging system.
No	GO to M2 .

M2 CHECK BATTERY CONDITION

- Ignition OFF.
- Refer to Section 414-01 and carry out the Battery — Condition Test.

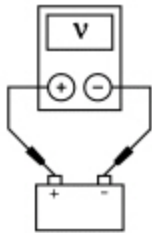
Does the battery pass the condition test?

Yes	GO to M3 .
No	CHARGE or INSTALL a new battery. REFER to Section 414-01. CLEAR all <u>CMDTCs</u> . TEST the system for normal operation.

M3 CHECK THE CHARGING SYSTEM VOLTAGE

NOTE: Do not allow the engine rpm to increase above 2,000 rpm while performing this step or the generator may self excite and result in default charging system output voltage. If engine rpm has gone above 2,000 rpm, shut the vehicle OFF and restart the engine before performing this step.

- Start the engine.
- Measure the voltage of the battery:
 - For DTC P0562:00, turn on headlights and HVAC fan on high and run engine at 1,500 rpm for a minimum of 2 minutes.
 - For DTC P0563:00, turn off all accessories and run the engine at 1,500 rpm for a minimum of 2 minutes.



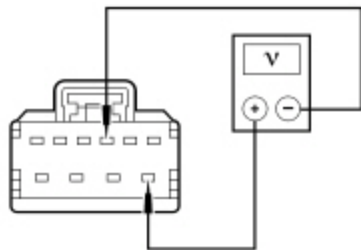
AJ0210-A

Is the battery voltage between 13-15.2 volts?

Yes	For DTC P0562:00, GO to M4 . For DTC P0563:00, GO to M6 .
No	REFER to Section 414-00 to diagnose the charging system. CLEAR all <u>CMDTCs</u> . TEST the system for normal operation.

M4 CHECK VOLTAGE AND GROUND TO THE TCCM

- Ignition OFF.
- Disconnect: TCCM [C2371B](#) .
- Ignition ON.
- Measure the voltage between TCCM [C2371B](#) Pin 3, circuit GD123 (BK/GY), harness side and TCCM [C2371B](#) Pin 7, circuit SBB68 (GN/RD), harness side.



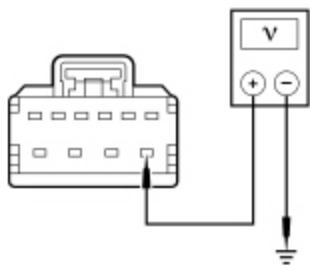
• N0091388

Is the voltage greater than 11 volts?

Yes	GO to M6 .
No	GO to M5 .

M5 CHECK VOLTAGE TO THE TCCM

- Measure the voltage between ~~TCCM~~ C2371B-7, circuit SBB68 (GN/RD), harness side and ground.



• N0091389

Is the voltage greater than 11 volts?

Yes	REPAIR circuit GD123 (BK/GY). CLEAR all CMDTCs . TEST the system for normal operation.
No	VERIFY BJB fuse 68 (20A) is OK. If OK, REPAIR circuit SBB68 (GN/RD). If not OK, REFER to the Wiring Diagrams manual to identify possible causes of the circuit short. CLEAR all CMDTCs. TEST the system for normal operation.

M6 CHECK FOR CORRECT TCCM OPERATION

- Ignition OFF.
- Disconnect and inspect all ~~TCCM~~ connectors.
- Repair:
 - corrosion (install new connector or terminals - clean module pins)
 - damaged or bent pins - install new terminals/pins
 - pushed-out pins - install new pins as necessary
- Connect all ~~TCCM~~ connectors. Make sure they are seated correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new TCCM . REFER to Transfer Case Control Module (TCCM) in this section. TEST the system for normal operation.
No	The system is operating correctly at this time. Concern may have been caused by a loose or corroded connector. ADDRESS the root cause of the connector or pin issues.

Pinpoint Test N: The Front Axle Does Not Engage/Disengage Correctly

Refer to Wiring Diagrams Cell [34](#) , Four-Wheel Drive Systems for schematic and connector information.

Locator views for the vacuum lines and vacuum reservoir are located in Description and Operation. Refer to Section 308-07A.

Normal Operation

In 2H mode, the **TCCM** supplies a ground path to the **IWE** solenoid to apply vacuum to the **IWEs** (disengaging the front hubs from the front axle). In 4H or 4L modes, the **TCCM** does not supply the ground path to the **IWE** solenoid, vacuum is not applied to **IWEs** the and an internal spring in the **IWEs** keeps the hubs engaged.

- DTC C1979:11 (IWE Solenoid Circuit Failure: Short Circuit To Ground) — Fault sets when **TCCM** C2371A pin 6 detects a short to ground. Fault results in **IWEs** hubs disengaged all the time.
- DTC C1979:12 (IWE Solenoid Circuit Failure: Short Circuit To Battery) — Fault sets when **TCCM** C2371A pin 6 detects a short to voltage. Fault results in **IWEs** hubs engaged all the time.
- DTC C1979:13 (IWE Solenoid Circuit Failure: Circuit Open) — Fault sets when **TCCM** C2371A pin 6 detects an open circuit. Fault results in **IWEs** hubs engaged all the time.

This pinpoint test is intended to diagnose the following:

- **MSS**
- **IWEs**
- **IWE** solenoid
- Transfer case
- Vacuum leaks
- Front axle assembly
- **TCCM**
- Wiring, terminals or connectors
- Fuses

PINPOINT TEST N : THE FRONT AXLE DOES NOT ENGAGE/DISENGAGE CORRECTLY

WARNING: When directed to drive the vehicle as part of this test, drive the vehicle on a hard surface in an area without traffic to prevent a crash. Failure to follow these instructions may result in personal injury.

N1 CHECK FOR TCCM DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self-Test — **TCCM** .

Are any DTCs present?

Yes	For DTC C1979:11, GO to N19 . For DTC C1979:12, GO to N18 .
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	For DTC C1979:13, GO to N20 . For all other DTCs, REFER to the TCCM - DTC Chart.
No	GO to N2 .

N2 CHECK FOR GRINDING NOISES AT HIGH SPEEDS

- With the vehicle warm and the MSS in 2H mode, drive the vehicle at sustained WOT for 5 seconds.

Is there a grinding noise from the axle system during WOT ?

Yes	GO to Pinpoint Test T.
No	GO to N3 .

N3 CHECK TCCM OPERATION

- Ignition ON.
- Monitor the 4X4 indicator on the IPC while moving the MSS between 2H and 4H modes.

Does the 4X4 indicator illuminate when the MSS is in the 4H position and turn off when the MSS is in the 2H position?

Yes	GO to N4 .
No	GO to Pinpoint Test X.

N4 CHECK THE IWE DISENGAGEMENT

NOTE: The engine must be idling during the following steps to supply the vacuum required for the IWEs to engage and disengage.

- With the vehicle in NEUTRAL, position it on a hoist. Refer to Section 100-02.
- Place the MSS in 2H.
- Start the engine.
- Rotate the LH front tire one revolution forward and one revolution backward while observing the LH front halfshaft.
- Rotate the RH front tire one revolution forward and one revolution backward while observing the RH front halfshaft.

Did either front halfshaft rotate?

Yes	GO to N8 .
No	GO to N5 .

N5 CHECK THE IWE ENGAGEMENT

- With the vehicle in NEUTRAL, position it on a hoist. Refer to Section 100-02.
- Place the MSS in 4H.
- Start the engine.
- Rotate the LH front tire one revolution forward and one revolution backward while observing the LH front halfshaft.
- Rotate the RH front tire one revolution forward and one revolution backward while observing the RH front halfshaft.

Did both front halfshafts rotate?

Yes	The front axle is operating properly at this time. CARRY OUT the 4WD Functional Test. GO to Pinpoint Test U.
No	GO to N6 .

N6 CHECK THE IWE SOLENOID OUTPUT IN 4H

- Disconnect the output vacuum line at the IWE solenoid.
- Switch the MSS to 4H.
- With the engine running, measure the vacuum at the output port of the IWE solenoid.

Is the vacuum less than 25 mm (1 in) Hg?

Yes	The IWE is mechanically stuck in the disengaged position. INSTALL a new IWE on the halfshaft(s) that did not rotate. REFER to Integrated Wheel End (IWE) in this section. TEST the system for normal operation.
No	GO to N7 .

N7 CHECK THE IWE SOLENOID

- Ignition OFF.
- Disconnect: IWE Solenoid .
- Start the engine.
- Measure the vacuum at the output port of the IWE solenoid.

Is the vacuum less than 25 mm (1 in) Hg?

Yes	GO to N20 .
No	INSTALL a new IWE solenoid. TEST the system for normal operation.

N8 CHECK THE IWE SOLENOID OUTPUT VACUUM IN 2H

- Disconnect the output vacuum line at the IWE solenoid.

- With the engine running and the **MSS** in 2H, measure the vacuum at the output port of the **IWE** solenoid.

Is the vacuum greater than 254 mm (10 in) Hg?

Yes	GO to N11 .
No	GO to N9 .

N9 CHECK THE IWE SOLENOID INPUT VACUUM

- Disconnect the supply vacuum line at the **IWE** solenoid.
- With the engine running, measure the supply line vacuum to the **IWE** solenoid.

Is the vacuum greater than 254 mm (10 in) Hg?

Yes	GO to N20 .
No	GO to N10 .

N10 CHECK THE VACUUM RESERVOIR

- Ignition OFF.
- With the source vacuum line still disconnected, connect a vacuum pump to the vacuum port on the vacuum reservoir and apply 508 mm (20 in) Hg of vacuum.

Does the vacuum maintain or drop less than 25 mm (1 in) Hg per minute?

Yes	REPAIR or INSTALL new vacuum lines or INSTALL a check valve. TEST the system for normal operation.
No	INSTALL a new vacuum reservoir. TEST the system for normal operation.

N11 CHECK THE IWE VACUUM LINES

- Ignition OFF.
- Disconnect the output vacuum line from the **IWE** solenoid.
- Disconnect and plug the LH and RH **IWE** vacuum lines at the **IWEs** .
- Connect a vacuum pump to the **IWE** output vacuum circuit at the **IWE** solenoid connection. Apply 508 mm (20 in) Hg of vacuum and observe the vacuum reading.

Does the vacuum drop more than 25 mm (1 in) Hg per minute?

Yes	GO to N14 .
No	GO to N12 .

N12 CHECK THE LH IWE VACUUM LINE FOR AN OBSTRUCTION

- REMOVE the plug from the vacuum line at the LH IWE .
- Connect a hand vacuum pump to the LH IWE vacuum line and attempt to draw a vacuum on the line.

Did the vacuum pump draw a vacuum on the LH IWE vacuum line?

Yes	INSTALL a new vacuum line. CLEAR the DTCs. TEST the system for normal operation.
No	GO to N13 .

N13 CHECK THE RH IWE VACUUM LINE FOR AN OBSTRUCTION

- REMOVE the plug from the vacuum line at the RH IWE .
- Connect a vacuum pump to the RH IWE vacuum line and attempt to draw a vacuum on the line.

Did the vacuum pump draw a vacuum on the RH IWE vacuum line?

Yes	INSTALL a new vacuum line. CLEAR the DTCs. TEST the system for normal operation.
No	GO to N14 .

N14 CHECK THE RH IWE LEAK RATE

- Apply 508 mm (20 in) Hg of vacuum to the RH IWE port and observe vacuum reading.

Does the vacuum maintain or drop less than 25 mm (1 in) Hg per minute?

Yes	GO to N15 .
No	INSTALL a new RH IWE . REFER to Integrated Wheel End (IWE) in this section. TEST the system for normal operation.

N15 CHECK THE RH IWE OPERATION

- Connect a vacuum pump with gauge to the larger port of the RH IWE .
- Observe the RH halfshaft while rotating the RH front tire. Apply between 152 and 203 mm (6 to 8 in) Hg of vacuum and release the vacuum to manually disengage and engage the RH IWE .

Did the RH IWE disengage between 152 and 203 mm (6 to 8 in) Hg and engage when the vacuum is released?

Yes	CONNECT the vacuum line to the RH IWE . GO to N16 .
No	INSTALL a new RH IWE . REFER to Integrated Wheel End (IWE) in this section. TEST the system for normal operation.

N16 CHECK THE LH IWE LEAK RATE

- Disconnect the vacuum line from the larger port at the LH IWE .

- Apply 508 mm (20 in) Hg of vacuum to the LH IWE port and observe vacuum reading.

Does the vacuum maintain or drop less than 25 mm (1 in) Hg per minute?

Yes	GO to N17 .
No	INSTALL a new LH IWE . REFER to Integrated Wheel End (IWE) in this section. TEST the system for normal operation.

N17 CHECK THE LH IWE OPERATION

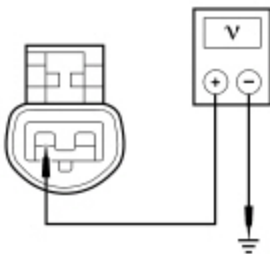
- Connect a vacuum pump with gauge to the larger port of the LH IWE .
- Observe the LH halfshaft while rotating the LH front tire. Apply between 152 and 203 mm (6 to 8 in) Hg of vacuum and release the vacuum to manually disengage and engage the LH IWE .

Did the LH IWE disengage between 152 and 203 mm (6 to 8 in) Hg and engage when the vacuum is released?

Yes	CONNECT the vacuum line to the LH IWE . GO to N18 .
No	INSTALL a new LH IWE . REFER to Integrated Wheel End (IWE) in this section. CLEAR the DTCs. TEST the system for normal operation.

N18 CHECK THE IWE SOLENOID CONTROL CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: IWE Solenoid .
- Disconnect: TCCM [C2371A](#) .
- Ignition ON.
- Measure the voltage between IWE solenoid [C1291](#) Pin 2, circuit CCF05 (BN/BU), harness side and ground.



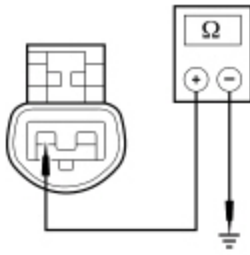
• N0091392

Is any voltage present?

Yes	REPAIR circuit CCF05 (BN/BU). CLEAR the DTCs. TEST the system for normal operation.
No	GO to N19 .

N19 CHECK THE IWE SOLENOID CONTROL CIRCUIT FOR A SHORT TO GROUND

- Ignition OFF.
- Disconnect: **IWE** Solenoid .
- Disconnect: **TCCM** [C2371A](#) .
- Measure the resistance between solenoid [C1291](#) Pin 2, circuit CCF05 (BN/BU), harness side and ground.



• N0091394

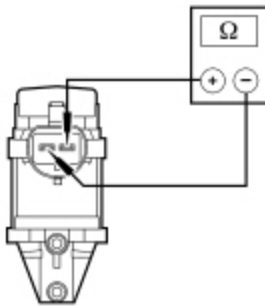
Is the resistance greater than 10,000 ohms?

Yes	GO to N20 .
No	REPAIR circuit CCF05 (BN/BU). CLEAR the DTCs. TEST the system for normal operation.

N20 CHECK THE IWE SOLENOID RESISTANCE

NOTE: *If the temperatures (either ambient or in the engine compartment) are exceptionally high, a higher than normal resistance reading could result. A lower temperature could result in a lower resistance reading.*

- Measure the resistance between **IWE** solenoid C1291 pins 1 and 2, component side.



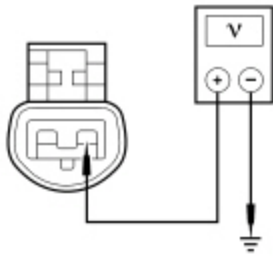
• N0091395

Is the resistance between 44 and 70 ohms?

Yes	GO to N21 .
No	INSTALL a new IWE solenoid. CLEAR the DTCs. TEST the system for normal operation.

N21 CHECK FOR VOLTAGE TO THE IWE SOLENOID

- Measure the voltage between **IWE** solenoid [C1291](#) Pin 1, circuit SBB29 (GY/RD), harness side and ground.



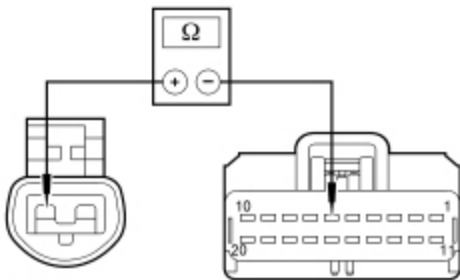
• N0091391

Is the voltage greater than 11 volts?

Yes	GO to N22 .
No	CHECK BJB fuse 29 (10A). If OK, REPAIR circuit SBB29 (GY/RD). If not OK, REFER to the Wiring Diagrams manual to identify possible causes of the circuit short. CLEAR the DTCs. TEST the system for normal operation.

N22 CHECK THE IWE SOLENOID CONTROL CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: ~~IWE~~ Solenoid .
- Disconnect: ~~TCCM~~ [C2371A](#) .
- Measure the resistance between ~~IWE~~ solenoid [C1291](#) Pin 2, circuit CCF05 (BN/BU), harness side and ~~TCCM~~ [C2371A](#) Pin 6, circuit CCF05 (BN/BU), harness side.



• N0091393

Is the resistance less than 3 ohms?

Yes	GO to N23 .
No	REPAIR circuit CCF05 (BN/BU). CLEAR the DTCs. TEST the system for normal operation.

N23 CHECK FOR CORRECT TCCM OPERATION

- Ignition OFF.
- Disconnect and inspect all ~~TCCM~~ connectors.
- Repair:
 - corrosion (install new connector or terminals - clean module pins)
 - damaged or bent pins - install new terminals/pins
 - pushed-out pins - install new pins as necessary

- Connect all **TCCM** connectors. Make sure they are seated correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new TCCM . REFER to Transfer Case Control Module (TCCM) in this section. TEST the system for normal operation.
No	The system is operating correctly at this time. Concern may have been caused by a loose or corroded connector. ADDRESS the root cause of the connector or pin issues.

Pinpoint Test O: The Vehicle Does Not Shift Between 4A (4X4 AUTO) and 4H (4X4) Modes Correctly

Refer to Wiring Diagrams Cell [34](#) , Four-Wheel Drive Systems for schematic and connector information.

Normal Operation

In 2H mode, the **TCCM** supplies a ground path to the **IWE** solenoid to apply vacuum to the **IWEs** (disengaging the front hubs from the front axle). In AUTO, 4H and 4L modes, the **TCCM** does not supply the ground path to the **IWE** solenoid, vacuum is not applied to the **IWEs** and an internal spring keeps the hubs engaged. Systematically check the necessary inputs and outputs at the **TCCM** , internal components of the transfer case, **MSS** , **IWE** components and drive axles.

This pinpoint test is intended to diagnose the following:

- **MSS**
- **IWEs**
- **IWE** solenoid
- Transfer case
- Vacuum leaks
- Front axle assembly
- **TCCM**
- Wiring, terminals or connectors

PINPOINT TEST O : THE VEHICLE DOES NOT SHIFT BETWEEN 4A (4X4 AUTO) AND 4H (4X4) MODES CORRECTLY

O1 CHECK FOR TCCM DTCS	
• Ignition ON. • Enter the following diagnostic mode on the scan tool: Self-Test — TCCM .	
Are DTCs retrieved?	
Yes	REFER to the TCCM DTC Chart in this section.
No	GO to O2 .
O2 CHECK THE 4H INDICATOR	
• Select 4H mode and observe the IPC .	

Does the IPC indicator lamp indicate 4H (4X4) mode?

Yes	GO to O8 .
No	GO to O3 .

O3 CHECK THE TCCM IGNITION POSITION (KEY_POS_4X4) PID

- Enter the following diagnostic mode on the scan tool: DataLogger — TCCM .
- Monitor the KEY_POS_4X4 PID while cycling through each position:

Does the ignition position PID match the ignition switch positions?

Yes	GO to O4 .
No	REFER to Section 211-05 to diagnose the ignition switch concern.

O4 CHECK THE TCCM 4WD SWITCH STATUS (4X4_SW) PID

- Monitor 4X4_SW PID while cycling the MSS through each position.

MSS Position	PID Status
2H	2-W Drive
4A	A4WD
4H	4W Auto/Hi
4L	4W Low

Does each PID status agree with each MSS position?

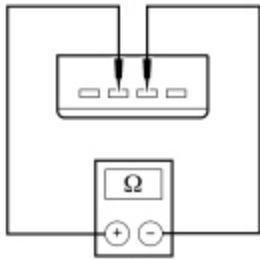
Yes	GO to O7 .
No	GO to O5 .

O5 CHECK THE MSS RESISTANCE VALUES

- Ignition OFF.
- Disconnect: MSS .
- Measure the resistance between C284 pin 2 and pin 3, component side. Refer to the following chart:

MSS Position	Resistance (±1 %)
2H	619 ohms
4A	270 ohms

MSS Position	Resistance ($\pm 1\%$)
4H	130 ohms
4L	62 ohms



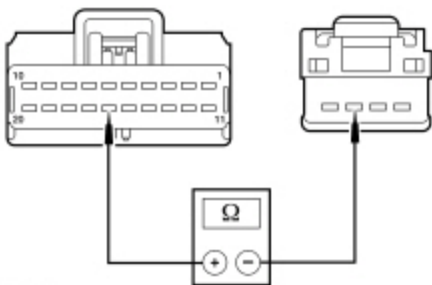
• N0063005

Are the resistance values within specification?

Yes	GO to O6 .
No	INSTALL a new MSS . REFER to Mode Select Switch (MSS) in this section. CLEAR the DTCs. TEST the system for normal operation.

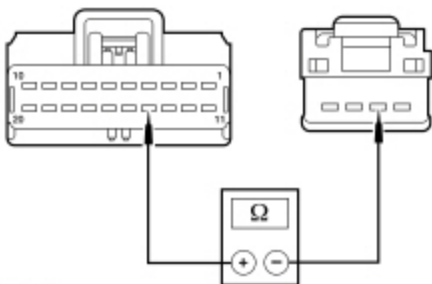
O6 CHECK THE MSS CONTROL AND RETURN CIRCUITS FOR AN OPEN

- Disconnect: TCCM C2371A .
- Measure the resistance between MSS [C284](#) Pin 2, circuit CCF13 (WH/BU), harness side and TCCM [C2371A](#) Pin 16, circuit CCF13 (WH/BU), harness side.



• N0101892

- Measure the resistance between MSS [C284](#) Pin 3, circuit RCF13 (WH/BN), harness side and TCCM [C2371A](#) Pin 14, circuit RCF13 (WH/BN), harness side.



• N0101893

Are the resistances less than 3 ohms?

Yes	GO to O19 .
No	REPAIR the affected circuit(s). CLEAR the DTCs. TEST the system for normal operation.

O7 CHECK THE TCCM 4WD PWM CLUTCH STATUS (4WD_CLTCH_OUT) PID IN 4H

- Switch the ~~MSS~~ to 4H mode.
- Monitor the 4WD_CLTCH_OUT PID.

Does the PID read 2 to 10% at closed throttle?

Yes	GO to O19 .
No	DIAGNOSE the PCM. REFER to PC/ED manual and the HS-CAN . REFER to Section 418-00. CLEAR the DTCs. TEST the system for normal operation.

O8 CHECK FOR WIND-UP IN 4H MODE

- Switch the ~~MSS~~ to 4H mode.
- Drive the vehicle on a dry, hard surface in turns while applying the throttle.

Is wind-up present in turns?

Yes	GO to O13 .
No	GO to O9 .

O9 CHECK THE TCCM 4WD PWM CLUTCH STATUS (4WD_CLTCH_OUT) PID AT WOT

- Enter the following diagnostic mode on the scan tool: DataLogger — ~~TCCM~~ .
- Monitor the 4WD_CLTCH_OUT PID at ~~WOT~~ , ~~KOEO~~ .

Does the 4WD_CLTCH_OUT PID read 98% at ~~WOT~~ ?

Yes	GO to O11 .
No	GO to O10 .

O10 CHECK THE TCCM ABSOLUTE THROTTLE POSITION (TP_TCCM) PID

- **NOTE:** The TP_TCCM PID is ~~APP~~, as communicated by the PCM via ~~HS-CAN~~ .

Slowly move the accelerator pedal from closed throttle to ~~WOT~~ and observe the TP_TCCM PID.

Does the TP_TCCM PID vary between 0% at closed throttle and 100% at ~~WOT~~ ?

Yes	GO to O11 .
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No	DIAGNOSE the PCM. REFER to PC/ED manual and the HS-CAN . REFER to Section 418-00. CLEAR the DTCs. TEST the system for normal operation.
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O11 CHECK FOR TRANSFER CASE CLUTCH ENGAGEMENT WITH THE TCCM 4WD PWM CLUTCH STATUS COMMAND (4WD_CLTCH_OUT) ACTIVE COMMAND AND 4WD CLUTCH STATUS (4WD_CLTCH_OUT) PIDS

- Using the 4WD_CLTCH_OUT active command, engage the transfer case clutch to 100% while monitoring the 4WD_CLTCH_OUT PID.

Does the 4WD_CLTCH_OUT PID read 100% ?

Yes	END the active command. CHECK the transfer case. REFER to Section 308-07B.
No	END the active command. GO to O12 .

O12 CHECK THE IPC INDICATOR IN AUTO MODE

- Select 4A mode and observe the IPC .

Does the message center indicate 4A (4X4 AUTO) mode?

Yes	GO to O13 .
No	GO to Pinpoint Test V .

O13 CHECK FOR WIND-UP IN AUTO MODE

- Drive the vehicle on a dry, hard surface in turns while applying the throttle.

Is wind-up present in turns?

Yes	GO to O14 .
No	The system is operating correctly.

O14 CHECK THE TCCM 4WD PWM CLUTCH STATUS (4WD_CLTCH_OUT) PID AT CLOSED THROTTLE

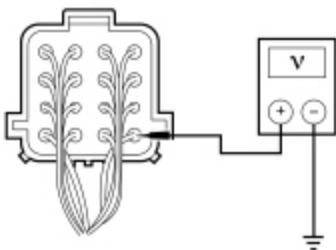
- Monitor the 4WD_CLTCH_OUT PID at closed throttle, KOEO .

Does the 4WD_CLTCH_OUT PID indicate 2 to 10% clutch application at closed throttle?

Yes	GO to O15 .
No	GO to O17 .

O15 CHECK VOLTAGE ON THE CLUTCH CONTROL CIRCUIT USING THE TCCM 4WD PWM CLUTCH STATUS (4WD_CLTCH_OUT) ACTIVE COMMAND

- Using the 4WD_CLTCH_OUT active command, command the transfer case clutch to 4%.
- Measure the voltage by carefully back-probing transfer case [C350](#) Pin 16, circuit CCF03 (GY/BN), component side and ground.



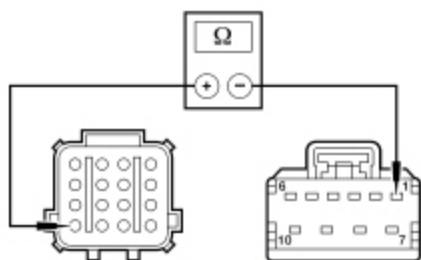
• N0133480

Is the voltage approximately 0.2 to 0.4 volt?

Yes	END the active command. CHECK the transfer case. REFER to Section 308-07B.
No	GO to O16 .

O16 CHECK THE CLUTCH CONTROL CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: Transfer Case .
- Disconnect: TCCM [C2371B](#) .
- Measure the resistance between transfer case [C350](#) Pin 16, circuit CCF03 (GY/BN), harness side and TCCM [C2371B](#) Pin 1, circuit CCF03 (GY/BN), harness side.



• N0093125

Are the resistances less than 3 ohms?

Yes	GO to O17 .
No	REPAIR circuit CCF03 (GY/BN). CLEAR the DTCs. TEST the system for normal operation.

O17 CHECK THE ABS MODULE WHEEL SPEED SENSOR PIDS

- Enter the following diagnostic mode on the scan tool: DataLogger — ABS Module .
- While driving the vehicle at 30 km/h (18 mph), monitor the following wheel speed sensor PIDs:
 - Left Front Wheel Speed Sensor (LF_WSPD)
 - Left Rear Wheel Speed Sensor (LR_WSPD)

- Right Front Wheel Speed Sensor (RF_WSPD)
- Right Rear Wheel Speed Sensor (RR_WSPD)

- Compare the speedometer reading to the wheel speed sensor PIDs.

Do the wheel speed sensor PIDs and speedometer speeds match within 1.5 km/h (1 mph)?

Yes	GO to O18 .
No	CHECK the ABS for faults. REFER to Section 206-09.

O18 CHECK THE TCCM ABSOLUTE THROTTLE POSITION (TP_TCCM) PID

- Enter the following diagnostic mode on the scan tool: DataLogger — ~~TCCM~~ .
- **NOTE:** *The TP_TCCM PID is ~~APP~~, as communicated by the PCM via ~~HS-CAN~~ .*

Slowly move the accelerator pedal from closed throttle to and observe the TP_TCCM PID.

Does the TP_TCCM PID vary between 0% at closed throttle and 100% at ~~WOT~~ ?

Yes	GO to O19 .
No	DIAGNOSE the PCM. REFER to PC/ED manual and the HS-CAN . REFER to Section 418-00. CLEAR the DTCs. TEST the system for normal operation.

O19 CHECK FOR CORRECT TCCM OPERATION

- Ignition OFF.
- Disconnect and inspect all ~~TCCM~~ connectors.
- Repair:
 - corrosion (install new connector or terminals - clean module pins)
 - damaged or bent pins - install new terminals/pins
 - pushed-out pins - install new pins as necessary
- Connect all ~~TCCM~~ connectors. Make sure they are seated correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new TCCM . REFER to Transfer Case Control Module (TCCM) in this section. TEST the system for normal operation.
No	The system is operating correctly at this time. Concern may have been caused by a loose or corroded connector. ADDRESS the root cause of the connector or pin issues.

Pinpoint Test P: The Vehicle Does Not Shift Between 4H (4X4) and 4L (4X4 Low) Modes Correctly

Refer to Wiring Diagrams Cell [34](#) , Four-Wheel Drive Systems for schematic and connector information.

Normal Operation

The high-low shift occurs when the reduction shift fork moves the high-low collar to lock the planetary gear set to the output shaft. The torque transmitted through the sun gear from the input shaft turns the front planetary gear set assembly. The front planetary gear set assembly, now engaged, provides transfer case speed reduction. Certain criteria, such as vehicle speed and TR selection must be met before this shift can occur. Systematically check the necessary inputs and outputs of the TCCM , components of the transfer case, IWE components and halfshafts.

DTC Description	Fault Trigger Condition
• P1839:00 — Transfer Case Shift Motor Open Circuit: No Sub Type Information	Fault sets when TCCM C2371B pin 4 or 5 detects an open circuit. Shift motor moves are prevented when this DTC is set.
• P1840:00 — Transfer Case Shift Motor Short Circuit To Battery: No Sub Type Information	Fault sets when TCCM C2371B pin 4 or 5 detects a short to voltage. Shift motor moves are prevented when this DTC is set.
• P1841:00 — Transfer Case Shift Motor Short Circuit To Ground: No Sub Type Information	Fault sets when TCCM C2371B pin 4 or 5 detects a short to ground. Shift motor moves are prevented when this DTC is set.
• P1849:00 — Transfer Case Contact Plate 'A' Short Circuit To Ground: No Sub Type Information	Fault sets when TCCM C2371A pin 1 detects a short to ground. Shift motor moves are prevented when this DTC is set. The TCCM will not complete mode change if a change in shift motor position is required.
• P184B:00 — Transfer Case Unable to Transition Between 4H and 4L: No Sub Type Information	Fault sets when target shift motor position is not reached after the maximum number of allowed blocked shift retries for 4H to 4L. Requested shift motor position is not reached.
• P1853:00 — Transfer Case Contact Plate 'B' Short Circuit to Ground: No Sub Type Information	Fault sets when TCCM C2371A pin 2 detects a short to ground. Shift motor moves are prevented when this DTC is set. The TCCM will not complete mode change if a change in shift motor position is required.
• P1857:00 — Transfer Case Contact Plate 'C' Short Circuit to Ground: No Sub Type Information	Fault sets when TCCM C2371A pin 3 detects a short to ground. Shift motor moves are prevented when this DTC is set. The TCCM will not complete mode change if a change in shift motor position is required.
• P1861:00 — Transfer Case Contact Plate 'D' Short Circuit to Ground: No Sub Type Information	Fault sets when TCCM C2371A pin 4 detects a short to ground. Shift motor moves are prevented when this DTC is set. The TCCM will not complete mode change if a change in shift motor position is required.

DTC Description	Fault Trigger Condition
P1867:00 — Transfer Case Contact Plate General Circuit Failure: No Sub Type Information	Fault sets when an invalid shift motor position is detected during selection of a new 4WD mode or during a motor movement. If the shift motor is not moving, shift motor movement will be prevented. If the shift motor is moving when this DTC sets, the shift motor will move to the high-side mechanical stop.
P1891:00 — Transfer Case Contact Plate Ground Return Open Circuit: No Sub Type Information	Fault sets when TCCM C2371A pin 5 detects a short to ground, a short to voltage or open circuit. The TCCM will not complete mode change if a change in shift motor position is required.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Transfer case
- ~~TR~~
- ABS wheel speed inputs
- ~~IWEs~~
- ~~TCCM~~
- Transmission drag torque

PINPOINT TEST P : THE VEHICLE DOES NOT SHIFT BETWEEN 4H (4X4) AND 4L (4X4 LOW) MODES CORRECTLY

WARNING: When directed to drive the vehicle as part of this test, drive the vehicle on a hard surface in an area without traffic to prevent a crash. Failure to follow these instructions may result in personal injury.

P1 CHECK FOR TCCM DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self-Test — ~~TCCM~~ .

Are any DTCs present?

Yes	For DTC P184B:00, GO to P2 . For DTC P1839:00, GO to P22 . For DTC P1840:00, GO to P23 . For DTC P1841:00, GO to P24 . For DTC P1849:00, P1853:00, P1857:00, P1861:00, GO to P17 . For DTC P1867:00, GO to P7 . For DTC P1891:00, GO to P15 . For all other DTCs, REFER to the TCCM DTC Chart in this section.
No	GO to P2 .

P2 CHECK THE 2H TO 4H SHIFT

- Drive the vehicle and shift ~~MSS~~ from 2H to 4H.

Does the vehicle shift between 2H and 4H correctly?

Yes	GO to P3 .
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No	GO to Pinpoint Test O.
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P3 CHECK THE 4H TO 4L SHIFT

- NOTE:** When switching in or out of 4L, the transmission must be in NEUTRAL and the vehicle stopped or moving less than 5 km/h (3 mph).

Drive the vehicle and shift MSS from 4H to 4L.

Does the transfer case shift from 4H to 4L correctly and the complaint is related to a clunk noise during the shift?

Yes	Transmission neutral drag is excessive. REFER to Section 307-01 for transmission diagnosis.
No	GO to P4 .

P4 CHECK THE NEUTRAL INDICATOR (NEUTRAL) PID

- Enter the following diagnostic mode on the scan tool: DataLogger — TCCM .
- Monitor the NEUTRAL TCCM PID while shifting the transmission through gear ranges.

Does the NEUTRAL PID indicate True when the transmission is shifted to NEUTRAL?

Yes	GO to P5 .
No	REFER to Section 307-01 for further diagnosis of the TR sensor.

P5 CHECK THE TCCM AVERAGE FRONT WHEEL SPEED (FRT_WHL_SPD) AND AVERAGE REAR WHEEL SPEED (REAR_WHL_SPD) PIDS

- Ignition OFF.
- Connect: TCCM [C2371A](#) .
- Connect: MSS .
- Ignition ON.
- Switch the MSS to 2H.
- Enter the following diagnostic mode on the scan tool: DataLogger — TCCM .
- Monitor the FRT_WHL_SPD and REAR_WHL_SPD PIDs with the vehicle stopped and while driving.

Do the FRT_WHL_SPD and REAR_WHL_SPD PIDs match with the vehicle stopped and at all speeds?

Yes	GO to P6 .
No	REFER to Section 206-09 to diagnose the wheel speed concern. After the repair is complete, CLEAR the DTCs. TEST the system for normal operation.

P6 CHECK THE ABS MODULE WHEEL SPEED PIDS

- Enter the following diagnostic mode on the scan tool: DataLogger — ABS Module .
- Monitor the wheel speed PIDs:
 - Left Front Wheel Speed Sensor (LF_WSPD)
 - Left Rear Wheel Speed Sensor (LR_WSPD)
 - Right Front Wheel Speed Sensor (RF_WSPD)
 - Right Rear Wheel Speed Sensor (RR_WSPD)

Do the wheel speed PIDs indicate 0 km/h (0 mph)?

Yes	GO to P7 .
No	REFER to Section 206-09 for further diagnosis of the ABS system.

P7 CHECK THE TCCM 4WD MODES (4WD_MODES) PID WITH THE MSS IN 4H

- Start the engine.
- Switch the MSS to 4H.
- Monitor the 4WD_MODES PID.

Does the 4WD_MODES PID indicate 4WD High?

Yes	GO to P10 .
No	GO to P8 .

P8 CHECK FOR 4H WIND-UP

- Switch the MSS to 4H.
- Drive the vehicle on a dry, hard surface in turns.

Is wind-up present in turns?

Yes	GO to P11 .
No	GO to P9 .

P9 CHECK FOR TRANSFER CASE ENGAGEMENT

- With the vehicle in NEUTRAL, position it on a hoist. Refer to Section 100-02.
- Rotate the rear driveshaft.
- Observe the front driveshaft.

Did the front driveshaft rotate?

Yes	GO to P10 .
No	GO to P14 .

P10 CHECK FOR IWE ENGAGEMENT

- With the vehicle in NEUTRAL, position it on a hoist. Refer to Section 100-02.
- Switch the MSS to 4H.
- Start the engine.
- **NOTE:** *The engine must be idling during the following steps to supply the vacuum required for the IWEs to engage and disengage.*

Rotate the LH front tire one revolution forward and one revolution backward while observing the LH front halfshaft and U-joint.

- Rotate the RH front tire one revolution forward and one revolution backward while observing the RH front halfshaft and U-joint.

Did both front halfshafts rotate?

Yes	GO to P12 .
No	GO to Pinpoint Test N .

P11 CHECK THE TRANSFER CASE SHIFT TO 2H

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: DataLogger — TCCM .
- Command the Transfer Case Contact Plate Power (PLATE_PWR_OS) PID to energize the transfer case gearmotor encoder assembly.
- Monitor the Transfer Case Contact Plate Switch A (PLATE_A), B (PLATE_B), C (PLATE_C) and D (PLATE_D) PIDs for the contact plate position with the MSS in the 2H position.

MSS Position	Contact Plate Position			
	PLATE_A	PLATE_B	PLATE_C	PLATE_D
2H/4A/4H Mode	CLOSED	OPEN	CLOSED	CLOSED
4L Mode	OPEN	CLOSED	OPEN	CLOSED

Do the PIDs and message center indicate that vehicle shifted into 2H (4X2 displayed in message center for approximately 4 seconds)?

Yes	GO to P12 .
No	GO to P15 .

P12 CHECK FOR IWE DISENGAGEMENT

- With the vehicle in NEUTRAL, position it on a hoist. Refer to Section 100-02.
- Start the engine.
- **NOTE:** *The engine must be idling during the following steps to supply the vacuum required for the IWEs to engage and disengage.*

Rotate the LH front tire one revolution forward and one revolution backward while observing the LH front halfshaft and U-joint.

- Rotate the RH front tire one revolution forward and one revolution backward while observing the RH front halfshaft and U-joint.

Did either of the front halfshafts rotate?

Yes	GO to Pinpoint Test N.
No	GO to P13 .

P13 CHECK FOR TRANSFER CASE DISENGAGEMENT

- Rotate the rear driveshaft while observing the front driveshaft.

Does the front driveshaft rotate?

Yes	The system is functioning correctly. CARRY OUT the Functional Test. GO to Pinpoint Test U.
No	GO to P14 .

P14 CHECK THE TCCM TRANSFER CASE CONTACT PLATE POSITION PIDS

- Enter the following diagnostic mode on the scan tool: DataLogger — TCCM .
- Command the Transfer Case Contact Plate Power (PLATE_PWR_OS) PID to energize the transfer case gearmotor encoder assembly.
- Monitor the Transfer Case Contact Plate Switch A (PLATE_A), B (PLATE_B), C (PLATE_C) and D (PLATE_D) PIDs for the contact plate position with the MSS in the 4H position.

MSS Position	Contact Plate Position			
	PLATE_A	PLATE_B	PLATE_C	PLATE_D
2H/4A/4H Mode	CLOSED	OPEN	CLOSED	CLOSED
4L Mode	OPEN	CLOSED	OPEN	CLOSED

Did the PIDs and message center indicate that the vehicle shifted into 4H (4X4 HIGH)?

Yes	GO to P25 .
No	GO to P15 .

P15 CHECK THE TRANSFER CASE SHIFT MOTOR OPERATION

- Remove the transfer case shift motor from the transfer case, leaving the wiring connector connected.
- Observe the transfer case shift motor while switching the **MSS** from 2H to 4H.

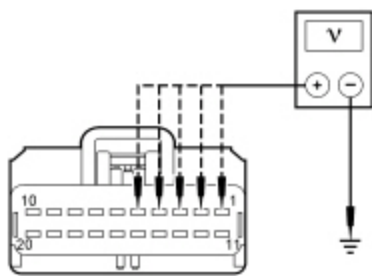
Does the transfer case shift motor rotate from the 2H position to the 4H position?

Yes	GO to P25 .
No	GO to P16 .

P16 CHECK TRANSFER PLATE CONTACT PLATE CIRCUITS FOR A SHORT TO VOLTAGE

- Ignition ON.
- Measure the voltage between the following circuits at [C2371A](#), harness side and ground:

Circuit	C2371A Pin
CCF14 (WH)	1
CCF15 (BN/YE)	2
CCF16 (VT/GN)	3
CCF17 (WH/OG)	4
RCF09 (BN/WH)	5



• N0091402

Is any voltage present?

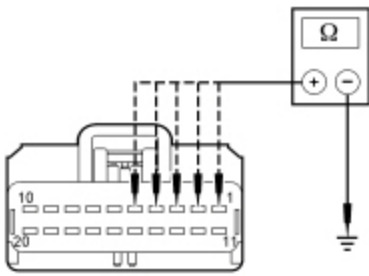
Yes	REPAIR the affected circuit(s). CLEAR the DTCs. TEST the system for normal operation.
No	GO to P17 .

P17 CHECK TRANSFER PLATE CONTACT PLATE CIRCUITS FOR A SHORT TO GROUND

- Ignition OFF.

- Disconnect: Transfer Case (if not previously disconnected) .
- Disconnect: TCCM [C2371A](#) (if not previously disconnected) .
- Measure the resistance between the following circuits at [C2371A](#), harness side and ground:

Circuit	C2371A Pin
CCF14 (WH)	1
CCF15 (BN/YE)	2
CCF16 (VT/GN)	3
CCF17 (WH/OG)	4
RCF09 (BN/WH)	5



N0091403

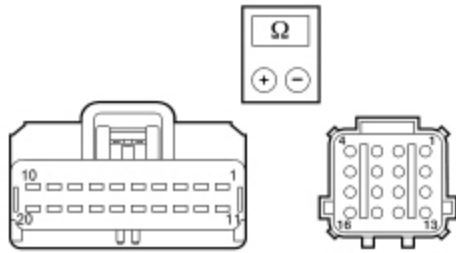
Are the resistances greater than 10,000 ohms?

Yes	GO to P18 .
No	REPAIR the affected circuit(s). CLEAR the DTCs. TEST the system for normal operation.

P18 CHECK THE TRANSFER PLATE CONTACT PLATE CIRCUITS FOR AN OPEN

- Ignition OFF.
- Disconnect: Transfer Case .
- Disconnect: TCCM [C2371A](#) .
- Measure the resistance of the following circuits between TCCM [C2371A](#), harness side and transfer case [C350](#), harness side:

Circuit	C2371A Pin	C350 Pin
CCF14 (GN/OG)	1	5
CCF15 (BN/YE)	2	1
CCF16 (VT/GN)	3	9
CCF17 (WH/OG)	4	13
RCF09 (BN/WH)	5	10



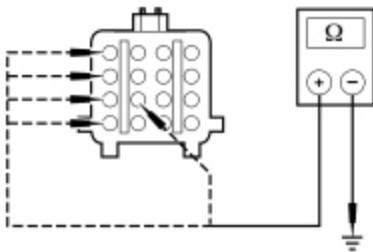
• N0091401

Are the resistances less than 3 ohms?

Yes	GO to P19 .
No	REPAIR the affected circuit(s). CLEAR the DTCs. TEST the system for normal operation.

P19 CHECK THE SHIFT MOTOR FOR AN INTERNAL SHORT TO GROUND

- Measure the resistance between transfer case C350 pins 1, 5, 9, 10 and 13, component side and ground.



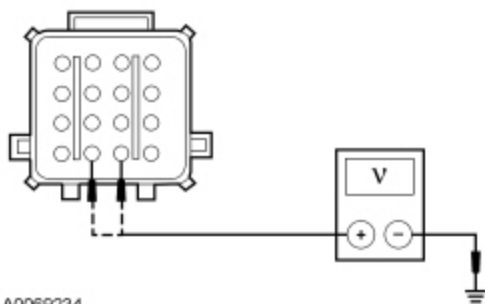
• N0091404

Are the resistances greater than 10,000 ohms?

Yes	GO to P20 .
No	INSTALL a new transfer case shift motor. REFER to Transfer Case Shift Motor in this section. CLEAR the DTCs. REPEAT the self-test. If the concern is still present, GO to P26 .

P20 CHECK THE SHIFT MOTOR CIRCUITS FOR POWER WHILE COMMANDING THE TCCM CW RELAY (CW_SMD_OS) AND CCW RELAY (CCW_SMD_OS) PIDS ON

- Connect: TCCM [C2371A](#) .
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: DataLogger — TCCM .
- Command the CW_SMD_OS PID ON and OFF while measuring the voltage between transfer case [C350](#) Pin 14, circuit CCF08 (GY/OG), harness side and ground. Then command the CCW_SMD_OS PID ON and OFF while measuring the voltage between transfer case [C350](#) Pin 15, circuit CCF07 (BN/GN), harness side and ground.



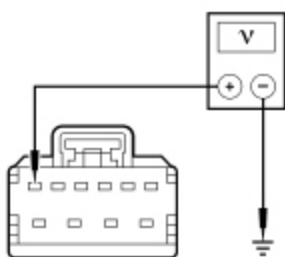
• A0069234

Are the voltages 10 volts or greater on the circuits when commanding the PIDs ON?

Yes	GO to P25 .
No	GO to P21 .

P21 CHECK THE TCCM VOLTAGE CIRCUIT

- Ignition OFF.
- Disconnect: TCCM [C2371B](#) .
- Measure the voltage between TCCM [C2371B](#) Pin 6, circuit SBB20 (GN/RD), harness side and ground.



• N0091405

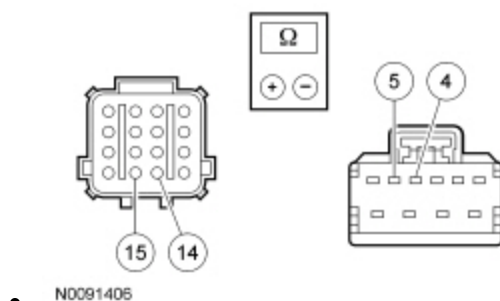
Is the voltage greater than 11 volts?

Yes	GO to P22 .
No	VERIFY BJB fuse 20 (20A) is OK. If OK, REPAIR circuit SBB20 (GN/RD). If not OK, REFER to the Wiring Diagrams manual to identify possible causes of the circuit short. CLEAR the DTCs. TEST the system for normal operation.

P22 CHECK THE TRANSFER CASE SHIFT MOTOR CIRCUITS FOR AN OPEN

- Ignition OFF.
- Disconnect: TCCM [C2371B](#) .
- Disconnect: Transfer Case .
- Measure the resistance between transfer case [C350](#), harness side and TCCM [C2371B](#), harness side. Refer to the following chart.

Transfer Case	Circuit	TCCM
C350 Pin 14	CCF08 (GY/OG)	C2371B Pin 5
C350 Pin 15	CCF07 (BN/GN)	C2371B Pin 4

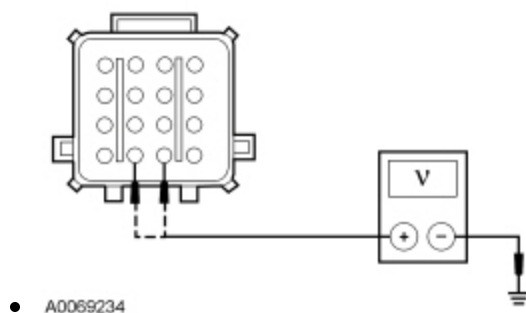


Are the resistances less than 3 ohms?

Yes	GO to P23 .
No	REPAIR the affected circuit(s). CLEAR the DTCs. TEST the system for normal operation.

P23 CHECK THE TRANSFER CASE SHIFT MOTOR CIRCUITS FOR SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: TCCM [C2371B](#) (if not previously disconnected) .
- Disconnect: Transfer Case (if not previously disconnected) .
- Measure the voltage between transfer case [C350](#) Pin 14, circuit CCF08 (GY/OG), harness side and ground; and between transfer case [C350](#) Pin 15, circuit CCF07 (BN/GN), harness side and ground.



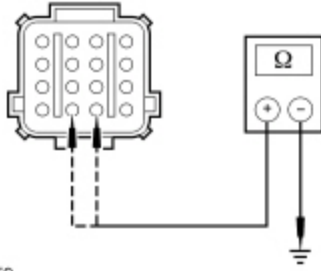
Is any voltage present?

Yes	REPAIR the affected circuit(s). CLEAR the DTCs. TEST the system for normal operation.
No	GO to P24 .

P24 CHECK THE TRANSFER CASE SHIFT MOTOR CIRCUITS FOR SHORT TO GROUND

- Ignition OFF.
- Disconnect: TCCM [C2371B](#) (if not previously disconnected) .

- Disconnect: Transfer Case (if not previously disconnected) .
- Measure the resistance between transfer case [C350](#) Pin 14, circuit CCF08 (GY/OG), harness side and ground; and between the transfer case [C350](#) Pin 15, circuit CCF07 (BN/GN), harness side and ground.



• N0055559

Are the resistances greater than 10,000 ohms?

Yes	GO to P25 .
No	REPAIR the affected circuit(s). CLEAR the DTCs. TEST the system for normal operation.

P25 CHECK FOR CORRECT TCCM OPERATION

- Ignition OFF.
- Disconnect and inspect all **TCCM** connectors.
- Repair:
 - corrosion (install new connector or terminals - clean module pins)
 - damaged or bent pins - install new terminals/pins
 - pushed-out pins - install new pins as necessary
- Connect all **TCCM** connectors. Make sure they are seated correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new TCCM . REFER to Transfer Case Control Module (TCCM) in this section. TEST the system for normal operation.
No	The system is operating correctly at this time. Concern may have been caused by a loose or corroded connector. ADDRESS the root cause of the connector or pin issues.

P26 SHIFT THE TRANSFER CASE TO 4H MANUALLY

- Remove the transfer case shift motor from the transfer case.
- Using a wrench, rotate the shift cam to the 4H position.
- Rotate the rear driveshaft and watch the front driveshaft.

Does the transfer case shift to 4H and the front driveshaft rotates?

Yes	GO to P27 .
No	REPAIR the transfer case. REFER to Section 308-07B.

P27 CHECK THE TORQUE REQUIRED TO SHIFT THE TRANSFER CASE

- Rotate the shift cam by hand through 4L, 4H and 2H positions.
- Measure the torque required to perform the shift.

Is the torque required to shift 45 Nm (33 lb-ft) or less?

Yes	GO to P28 .
No	REPAIR the transfer case. REFER to Section 308-07B.

P28 SHIFT THE TRANSFER CASE TO 2H MANUALLY

- Rotate the shift cam to the 2H position.
- Rotate the rear driveshaft and watch the front driveshaft.

Does the transfer case shift to 2H and the front driveshaft does not rotate?

Yes	GO to P29 .
No	REPAIR the transfer case. REFER to Section 308-07B.

P29 SHIFT THE TRANSFER CASE TO 4L MANUALLY

- Rotate the shift cam by hand to the 4L position.
- Rotate the rear driveshaft and watch the front driveshaft.

Does the transfer case shift to 4L and the front driveshaft rotates?

Yes	INSTALL a new transfer case shift motor. REFER to Transfer Case Shift Motor in this section. CLEAR the DTCs. TEST the system for normal operation.
No	REPAIR the transfer case. REFER to Section 308-07B.

Pinpoint Test Q: Four-Wheel Drive (4WD) Does Not Engage at Speed Correctly

Refer to Wiring Diagrams Cell [34](#) , Four-Wheel Drive System for schematic and connector information.

Normal Operation

When the operator selects a transfer case mode from 2H mode, the TCCM engages a clutch for approximately 5 seconds to synchronize the front and rear driveshafts allowing a mechanical engagement of the transfer case. The TCCM also removes

the ground path from the IWE solenoid, which removes vacuum from the IWEs , allowing them to engage the front axle shaft. Systematically check the necessary inputs and outputs at the TCCM , internal components of the transfer case, IWE components and drive axles.

- DTC P1825:00 (4-Wheel Drive Clutch Relay Open Circuit: No Sub Type Information) — Fault sets when TCCM C2371B pin 1 detects an open circuit. The TCCM disables transfer case synchronization clutch operation.
- DTC P1826:00 (4-Wheel Drive Low Clutch Relay Circuit To Battery: No Sub Type Information) — Fault sets when TCCM C2371B pin 1 detects a short to voltage. Transfer case synchronization clutch operation, though disabled by the TCCM , is fully engaged when this DTC is present.
- DTC P1827:00 (4-Wheel Drive Low Clutch Relay Circuit To Ground) — Fault sets when TCCM C2371B pin 1 detects a short to ground. The TCCM disables transfer case synchronization clutch operation.

This pinpoint test is intended to diagnose the following:

- Transfer case synchronization clutch
- Transfer case
- TCCM
- Front axle assembly
- Wiring, terminals or connectors

PINPOINT TEST Q : 4WD DOES NOT ENGAGE AT SPEED CORRECTLY

Q1 RETRIEVE TCCM DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self-Test — TCCM .

Are any DTCs retrieved?

Yes	For DTC P1825:00, GO to Q11 . For DTC P1826:00, GO to Q13 . For DTC P1827:00, GO to Q10 . For all other DTCs, GO to the TCCM DTC Chart in this section.
No	GO to Q2 .

Q2 CHECK 4X4 MODE ENGAGEMENT AT REST

- Switch the Mode Select Switch (MSS) from 4X2 mode to 4X4 mode.

Does 4X4 mode engage at rest correctly?

Yes	GO to Q3 .
No	GO to Pinpoint Test N .

Q3 CHECK 4X2 MODE TO 4X4 MODE SHIFT AT SPEED

- Switch the MSS to 4X2 mode.
- Drive the vehicle and carry out a 4X2 mode to 4X4 mode shift while driving at approximately 16 km/h (10 mph).
- Stop the vehicle, shift to 4X2 mode, drive the vehicle and carry out a 4X2 mode to 4X4 mode shift while driving at approximately 64 km/h (40 mph).

Does 4X4 mode engage correctly at 16 km/h (10 mph) and at 64 km/h (40 mph)?

Yes	GO to Pinpoint Test U.
No	GO to Q4 .

Q4 CHECK THE TCCM AVERAGE FRONT WHEEL SPEED (FRT_WHL_SPD) AND AVERAGE REAR WHEEL SPEED (REAR_WHL_SPD)

- Enter the following diagnostic mode on the scan tool: DataLogger — **TCCM** .
- Monitor the FRT_WHL_SPD and REAR_WHL_SPD PIDs while driving in a straight line.

Do the FRT_WHL_SPD and REAR_WHL_SPD PIDs match?

Yes	GO to Q5 .
No	REFER to Section 206-09 to diagnose the wheel speed sensor concern.

Q5 CHECK THE TRANSFER CASE CLUTCH LOCKUP USING TCCM 4WD PWM CLUTCH STATUS (4WD CLTCH_OUT) PID ACTIVE COMMAND

- With the vehicle in NEUTRAL, position it on a hoist. Refer to Section 100-02.
- Enter the following diagnostic mode on the scan tool: DataLogger — **TCCM** .
- Switch the **MSS** to AUTO mode.
- Using the 4WD CLTCH_OUT active command, command the PID to 98%.
- Rotate the rear driveshaft while observing the front driveshaft.

Does the front driveshaft rotate when the rear driveshaft is rotated?

Yes	GO to Q6 .
No	GO to Q9 .

Q6 CHECK THE TRANSFER CASE CLUTCH RELEASE USING TCCM 4WD PWM CLUTCH STATUS (4WD CLTCH_OUT) PID ACTIVE COMMAND

- With the vehicle still in NEUTRAL and on a hoist, using the 4WD_CLTCH_OUT active command, command the transfer case clutch to 0%.
- Rotate the rear driveshaft while observing the front driveshaft.

Does the front driveshaft rotate when the rear driveshaft is rotated?

Yes	GO to Q15 .
No	End the active command. GO to Q7 .

Q7 CHECK THE TRANSFER CASE ENGAGEMENT (ON A HOIST)

- Start the engine, switch the MSS to 4X2 mode.
- Place the transmission in DRIVE and increase speed to between 16 km/h (10 mph) and 24 km/h (15 mph).
- While observing the front driveshaft, switch the MSS to 4X4 mode.

Did the front driveshaft start to spin?

Yes	GO to Q8 .
No	GO to Q15 .

Q8 CHECK THE TRANSFER CASE DISENGAGEMENT

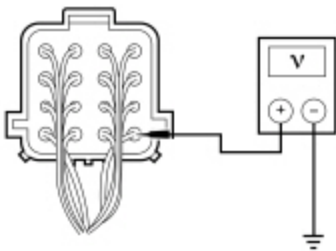
- Switch the MSS to 4X2 mode while observing the front driveshaft.

Does the front driveshaft rotate?

Yes	GO to Q15 .
No	GO to Pinpoint Test N .

Q9 CHECK CLUTCH CONTROL CIRCUIT FOR VOLTAGE USING THE TCCM 4WD PWM CLUTCH STATUS (4WD_CLTCH_OUT) ACTIVE COMMAND

- Using the 4WD_CLTCH_OUT active command, command the transfer case clutch to 98%.
- Measure the voltage by carefully back-probing transfer case [C350](#) Pin 16, circuit CCF03 (GY/BN), component side and the transfer case housing.



• N0133480

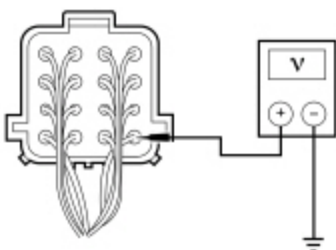
Is the voltage greater than 9 volts?

Yes	END the active command. INSTALL a new transfer case synchronization clutch. REFER to Section 308-07B. CLEAR the DTCs. TEST the system for normal operation.
No	GO to Q10 .

Q10 CHECK THE TRANSFER CASE-TO-CHASSIS GROUND USING THE TCCM 4WD PWM CLUTCH

STATUS (4WD_CLTCH_OUT) ACTIVE COMMAND

- Using the 4WD_CLTCH_OUT active command, command the transfer case clutch to 98%.
- Measure the voltage by carefully back-probing transfer case [C350](#) Pin 16, circuit CCF03 (GY/BN), component side and chassis ground.



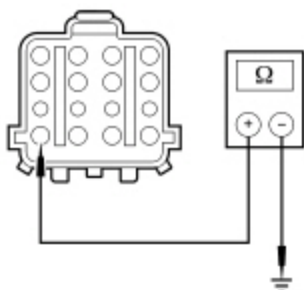
• N0133480

Is the voltage greater than 9 volts?

Yes	END the active command. REPAIR the transfer case ground connection between the transfer case and the chassis. CLEAR the DTCs. TEST the system for normal operation.
No	GO to Q11 .

Q11 CHECK CLUTCH CONTROL CIRCUIT FOR A SHORT TO GROUND

- Ignition OFF.
- Disconnect: Transfer Case .
- Disconnect: TCCM [C2371B](#) .
- Measure the resistance between transfer case [C350](#) Pin 16, circuit CCF03 (GY/BN), harness side and ground.



• N0057998

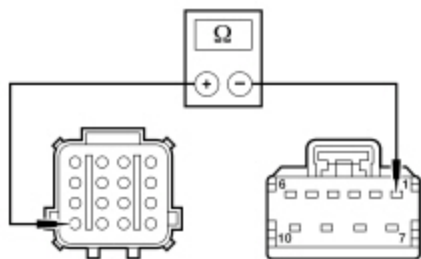
Is the resistance greater than 10,000 ohms?

Yes	GO to Q12 .
No	REPAIR circuit CCF03 (GY/BN). CLEAR the DTCs. TEST the system for normal operation.

Q12 CHECK CLUTCH CONTROL CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: Transfer Case (if not previously disconnected) .

- Disconnect: TCCM [C2371B](#) (if not previously disconnected) .
- Measure the resistance between transfer case [C350](#) Pin 16, circuit CCF03 (GY/BN), harness side and TCCM [C2371B](#) Pin 1, circuit CCF03 (GY/BN), harness side.



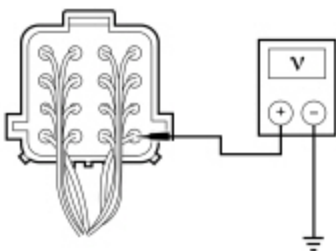
N0093126

Is the resistance less than 3 ohms?

Yes	GO to Q13 .
No	REPAIR circuit CCF03 (GY/BN). CLEAR the DTCs. TEST the system for normal operation.

Q13 CHECK CLUTCH CONTROL CIRCUIT FOR VOLTAGE USING THE TCCM 4WD PWM CLUTCH STATUS (4WD_CLTCH_OUT) PID ACTIVE COMMAND

- Connect: Transfer Case .
- Connect: TCCM [C2371B](#) .
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: DataLogger — TCCM .
- Using the 4WD_CLTCH_OUT active command, command the transfer case clutch to greater than 10%.
- Measure the voltage by carefully back-probing transfer case [C350](#) Pin 16, circuit CCF03 (GY/BN), component side and ground.



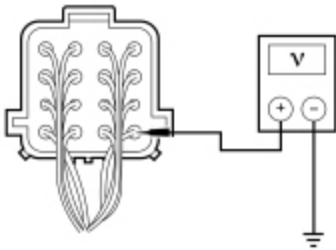
N0133480

Is any voltage present?

Yes	END the active command. GO to Q14 .
No	END the active command. INSTALL a new transfer case synchronization clutch. REFER to Section 308-07B. CLEAR the DTCs. TEST the system for normal operation.

Q14 CHECK CLUTCH CONTROL CIRCUIT FOR A SHORT TO VOLTAGE USING THE TCCM 4WD PWM CLUTCH STATUS (4WD_CLTCH_OUT) ACTIVE COMMAND

- Enter the following diagnostic mode on the scan tool: DataLogger — **TCCM** .
- Using the 4WD_CLTCH_OUT active command, command the transfer case clutch to 0%.
- Measure the voltage by carefully back-probing transfer case [C350](#) Pin 16, circuit CCF03 (GY/BN), component side and ground.



• N0133480

Is any voltage present?

Yes	REPAIR circuit CCF03 (GY/BN). CLEAR the DTCs. TEST the system for normal operation.
No	GO to Q15 .

Q15 CHECK FOR CORRECT TCCM OPERATION

- Ignition OFF.
- Disconnect and inspect all **TCCM** connectors.
- Repair:
 - corrosion (install new connector or terminals - clean module pins)
 - damaged or bent pins - install new terminals/pins
 - pushed-out pins - install new pins as necessary
- Connect all **TCCM** connectors. Make sure they are seated correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new TCCM . REFER to Transfer Case Control Module (TCCM) in this section. TEST the system for normal operation.
No	The system is operating correctly at this time. Concern may have been caused by a loose or corroded connector. ADDRESS the root cause of connector or pin issue.

Pinpoint Test R: The Four-Wheel Drive (4WD) System Jumps Out of Gear

Locator views for the vacuum lines and vacuum reservoir are located in Description and Operation. Refer to Section 308-07A.

Normal Operation

When the 4WD system engages the user selected mode, that mode should be maintained. If the 4WD system does not stay in the selected mode, systematically check the 4WD system components.

This pinpoint test is intended to diagnose the following:

- IWE vent line(s)
- MSS
- Transfer case

PINPOINT TEST R : THE 4WD SYSTEM JUMPS OUT OF GEAR

R1 CHECK FOR OBSTRUCTED IWE VENT LINES

- With the vehicle in NEUTRAL, position it on a hoist. Refer to Section 100-02.
- **NOTE:** *The IWE vent port is the smaller of the 2 vacuum ports.*

Disconnect the LH IWE line from the vent port.
- Attach a hand vacuum pump to the IWE vent line.
- Attempt to draw and hold a vacuum.
- Connect the LH IWE line.
- Disconnect the RH IWE line from the vent port.
- Attempt to draw and hold a vacuum.

Can a vacuum be held on either IWE vent line?

Yes	INSTALL new IWE vent lines as necessary. TEST the system for normal operation.
No	GO to R2 .

R2 CHECK THE TCCM 4WD SWITCH STATUS (4X4_SW) PID

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: DataLogger — TCCM .
- Drive the vehicle.
- Monitor the 4X4_SW PID while driving the vehicle in each MSS position.

MSS Position	PID Status
2H	2-W Drive
4A	A4WD
4H	4W Auto/Hi
4L	4W Low

Does the PID always agree with the MSS position?

Yes	GO to R3 .
No	GO to Pinpoint Test N .

R3 CHECK THE 4WD INDICATORS

- Stop the vehicle.
 - Shift the transmission into the NEUTRAL position.
 - **NOTE:** 2H is indicated by the absence of 4X4 AUTO, 4H or 4L mode indicators in the IPC message center.
- Observe the 4WD indicators in the IPC while cycling through each MSS position.

Do the 4WD indicators match MSS switch positions?

Yes	GO to Pinpoint Test N .
No	GO to R4 .

R4 CHECK FOR TCCM DTCS

- Shift the transmission into the PARK position.
- Enter the following diagnostic mode on the scan tool: Self-Test — TCCM .

Are DTCs present?

Yes	REFER to the TCCM DTC Chart in this section for diagnosis.
No	CHECK the transfer case fluid level. REFER to Section 308-07B. If the fluid level is OK, CHECK for internal transfer case faults. REFER to Section 308-07B. TEST the system for normal operation.

Pinpoint Test S: Vehicle Pulsates or Shudders in a Straight Line (Straightline Driveline Wind-up)

Normal Operation

In order for the 4WD system to function correctly, tires and wheels must be the same size, be in good condition and the front and rear axle ratios must match.

This pinpoint test is intended to diagnose the following:

- Unmatched tire sizes
- Unequal amounts of tire wear
- Unequal tire inflation pressures
- Unmatched front and rear axle ratios

PINPOINT TEST S : VEHICLE PULSATES OR SHUDDERS IN A STRAIGHT LINE (STRAIGHTLINE DRIVELINE WIND-UP)

NOTE: 4X4 is not intended for driving on hard/dry surfaces; using 4X4 mode while driving on these surfaces may produce noises like intermittent clunking, but will not damage driveline components.

S1 VERIFY THE CONCERN

- Drive the vehicle and shift from 2H mode to 4H mode. There should be minimal wind-up in 4H mode during straightline maneuvers (wind-up in turns is normal; 4H/4L is not intended for driving on hard/dry surfaces).

Is excessive wind-up present?

Yes	GO to S2 .
No	RETURN the vehicle to the customer. ADVISE about correct 4WD system operation and normal vehicle behavior.

S2 CHECK FOR MATCHING WHEEL AND TIRE SIZES

- Check the wheel and tire sizes.

Do the 4 wheels and tire sizes match?

Yes	GO to S3 .
No	ADVISE the customer that 4WD systems require 4 matched, correctly inflated, correctly maintained tires to operate correctly.

S3 CHECK TIRE WEAR

- Check each of the 4 tires for wear.

Are the 4 tires worn evenly?

Yes	GO to S4 .
No	ADVISE the customer that 4WD systems require 4 matched, correctly inflated, correctly rotated tires to operate correctly.

S4 CHECK TIRE INFLATION PRESSURE

- Check the inflation pressure in each tire. Refer to the Safety Certification Label on driver door or door pillar.

Are the tire inflation pressures correct?

Yes	GO to S5 .
No	ADJUST the tire pressure as necessary. TEST the vehicle for normal operation. ADVISE the customer that 4WD systems require 4 matched, correctly inflated, correctly maintained tires to operate correctly.

S5 CHECK THE FRONT AND REAR AXLE RATIOS

- Check that front and rear axle ratios match. Refer to Section 205-00.

Do the front and rear axle ratios match?

Yes	RETURN the vehicle to the customer and ADVISE about correct 4W.D system usage and normal vehicle behavior.
No	CHECK the V.C label. INSTALL the correct axle(s). REFER to Section 205-02A or Section 205-02B or Section 205-03. TEST the system for normal operation.

Pinpoint Test T: The Front Axle Makes Noise in 2H Mode Only and Generally Under Heavy Throttle

Refer to Wiring Diagrams Cell [34](#) , Four-Wheel Drive Systems for schematic and connector information.

Locator views for the vacuum lines and vacuum reservoir are located in Description and Operation. Refer to Section 308-07A.

Normal Operation

In 2H mode, the TCCM provides a ground path to the IWE solenoid to supply vacuum to each IWE (disengaging the front hubs from the front axle). During certain conditions such as WOT maneuvers, manifold vacuum is lower than desired. To maintain uninterrupted IWE function in 2H, stored engine vacuum is sourced from the vacuum reservoir. A one-way check valve is implemented between the vacuum reservoir and the IWE solenoid to prevent the stored vacuum from escaping into the manifold during low engine vacuum conditions. For noise-free IWE operation under all conditions, the check valve must be operating correctly.

When following this diagnostic procedure please note the following since these system interactions may influence the results:

- When shifting to 2H mode in order for the IWE to completely disengage the vehicle steering wheel must be less than 90 degrees and 5 seconds have elapsed since the shift to 2H has been initiated.
- The TCCM has a strategy which is used to warm up the front axle to improve driveline synchronization. When the gear selector is moved from park (P) to drive (D), the TCCM will turn off the IWE solenoid. The hubs stay engaged regardless of the selected 4x4 mode until a calibrated distance has been traveled above 6 mph (10 km/h). When ambient temperature is below 32°F (0°C), the distance is approximately 2 mi (3 km). When the ambient temperature is above 32° F (0°C), the distance is approximately 0.5 mi (800 m). Distance traveled resets when customer shifts to park. The warm up process will repeat when the gear selector is placed in drive (D).

This pinpoint test is intended to diagnose the following:

- Vacuum leaks
- Vacuum check valve
- Vacuum reservoir
- IWE solenoid
- IWE

PINPOINT TEST T : THE FRONT AXLE MAKES NOISE IN 2H MODE ONLY AND GENERALLY UNDER HEAVY THROTTLE

T1 CHECK FOR TCCM DTCS
WARNING: When directed to drive the vehicle as part of this test, drive the vehicle on a hard surface in an area without traffic to prevent a crash. Failure to follow these instructions may result in personal injury.

- Ignition ON
- Enter the following diagnostic mode on the scan tool: **TCCM KOEO Self Test.**

Are any DTC 's present?

Yes	REFER to the TCCM DTC chart.
No	GO to T2 .

T2 CHECK THE ONE-WAY VACUUM CHECK VALVE AND VACUUM RESERVOIR

NOTE: *The check valve is between the IWE solenoid and the vacuum reservoir (-19A563-).*

- Ignition OFF
- Disconnect the vacuum supply line elbow from the **IWE** solenoid.
- Connect a hand vacuum pump and apply 508 mm (20 in) Hg of vacuum to the check valve.

Does the vacuum drop less than 12.5 mm (0.5 in) Hg per minute?

Yes	The one-way check valve is holding vacuum. RECONNECT the vacuum supply line to the one-way vacuum check valve. GO to T4
No	Either the one-way check valve or vacuum reservoir is not holding vacuum.GO to T3

T3 CHECK THE VACUUM RESERVOIR

- Disconnect the vacuum line from the vacuum reservoir.
- Connect a hand vacuum pump and apply 508 mm (20 in) Hg of vacuum to the vacuum reservoir.

Does the vacuum drop less than 12.5 mm (0.5 in) Hg per minute?

Yes	The vacuum reservoir is holding vacuum. RECONNECT the vacuum supply line to the vacuum reservoir. GO to T4
No	The vacuum reservoir is not holding vacuum. INSTALL a new vacuum reservoir (-19A566-). TEST the system for normal operation. GO to Pinpoint Test U

T4 CHECK THE VACUUM SUPPLY LINE

- Disconnect the supply vacuum line at the **IWE** solenoid and allow the system to vent.
- Connect a hand vacuum pump and apply 508 mm (20 in) Hg of vacuum to the vacuum supply line, one-way check valve, and vacuum reservoir.

Does the vacuum drop less than 12.5 mm (0.5 in) Hg per minute?

Yes	The vacuum supply line is holding vacuum. RECONNECT the vacuum supply line to the IWE solenoid. GO to T5
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No	The vacuum supply line is not holding vacuum. INSTALL a new vacuum supply line and one-way check valve assembly (-7A788-). TEST the system for normal operation. GO to Pinpoint Test U
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T5 CHECK THE IWE SOLENOID OUTPUT VACUUM

- Ignition ON
- Enter the following diagnostic mode on the scan tool: **TCCM Datalogger**
- Command IWE_OS # to Yes (True)
- Disconnect the **IWE** solenoid output vacuum line at the **IWE** solenoid. The vacuum supply line remains connected to the **IWE** solenoid.
- Connect a vacuum gauge to the **IWE** solenoid output port.
- Start the engine and let it idle for 1 minute.

Is the vacuum greater than 254mm (10 in Hg) at the **IWE solenoid output port?**

Yes	The solenoid is operating correctly. Exit Output State Control. GO to T6 .
No	The IWE solenoid is blocked, restricted, or mechanically stuck closed. INSTALL a new IWE solenoid (-9H465-).

T6 CHECK THE IWE VACUUM SYSTEM

- Ignition OFF
- Disconnect the **IWE** solenoid output vacuum line at the **IWE** solenoid.
- Using a hand vacuum pump, apply 508 mm (20 in) Hg of vacuum to the **IWE** system through the solenoid output vacuum line.
- Observe the vacuum reading.

Does the vacuum drop less than 12.5 mm (0.5 in) Hg per minute?

Yes	The IWE vacuum system is not leaking. CONNECT the vacuum line to the solenoid. GO to T9 .
No	The IWE vacuum system is leaking. CONNECT the vacuum line to the solenoid. GO to T7 .

T7 CHECK THE RH IWE LEAK RATE

- With the vehicle in NEUTRAL, position it on a hoist. Refer to Section 100-02.
- Disconnect the vacuum line from the larger port on the RH **IWE**.
- Apply 508 mm (20 in) Hg of vacuum to the RH **IWE** port and observe vacuum reading.

Does the vacuum drop less than 12.5 mm (0.5 in) Hg per minute?

Yes	The RH IWE is not leaking. GO to T8 .
No	The RH IWE is leaking. INSTALL a new RH IWE . Integrated Wheel End (IWE) TEST the system for normal operation. GO to Pinpoint Test U .

T8 CHECK THE LH IWE LEAK RATE

- Disconnect the vacuum line from the LH IWE .
- Apply 508 mm (20 in) Hg of vacuum to the larger port on the LH IWE and observe vacuum reading.

Does the vacuum drop greater than 12.5 mm (0.5 in) Hg per minute?

Yes	Based on the results of E6, E7, and E8 the IWE system vacuum lines are leaking. INSTALL new vacuum lines (-7A785-, -3C125-) as necessary.
No	The LH IWE is leaking. INSTALL a new LH IWE . Integrated Wheel End (IWE) TEST the system for normal operation. GO to Pinpoint Test U

T9 CHECK THE RH IWE VACUUM LINE FOR AN OBSTRUCTION

- With the vehicle in NEUTRAL, position it on a hoist. Section 100-02
- Disconnect the vacuum line at the RH IWE .
- Connect a hand-held vacuum gauge to the larger vacuum supply line at the RH IWE .
- Ignition ON
- Enter the following diagnostic mode on the scan tool: ~~TCCM~~ Datalogger
- Command IWE_OS # to Yes (True)
- Start the engine and let it idle for 1 minute.

Is the vacuum greater than 254mm (10 in Hg) at the RH IWE ?

Yes	The vacuum line to the RH IWE is not blocked or restricted. Remove the vacuum gauge and CONNECT the vacuum lines to the RH IWE . GO to T10
No	The IWE system vacuum line is blocked between the IWE solenoid and the RH IWE . INSTALL new vacuum lines (-7A785-, -3C125-) as necessary.

T10 CHECK THE LH IWE VACUUM LINE FOR AN OBSTRUCTION

- Disconnect the vacuum line at the LH IWE .
- Connect a hand-held vacuum gauge to the larger vacuum supply line at the LH IWE .
- Ignition ON
- Enter the following diagnostic mode on the scan tool: ~~TCCM~~ Datalogger
- Command IWE_OS # to Yes (True)

- Start the engine and let it idle for 1 minute.

Is the vacuum greater than 254mm (10 in Hg) at the LH IWE ?

Yes	The IWE system vacuum lines are operating correctly at this time. GO to T11
No	The IWE system vacuum line is blocked between the IWE solenoid and the LH IWE . INSTALL new vacuum lines (-7A785-, -3C125-) as necessary.

T11 CHECK THE RH IWE OPERATION

- Ignition OFF
- Connect a hand vacuum pump with gauge to the larger port of the RH IWE .
- Observe the RH halfshaft while rotating the RH front tire. Apply between 152 and 203 mm (6 to 8 in) Hg of vacuum and release the vacuum to manually disengage and engage the RH IWE .

Did the RH IWE disengage between 152 and 203 mm (6 to 8 in) Hg and engage when the vacuum is released?

Yes	The RH IWE is operating correctly at this time. GO to T12
No	Remove the RH IWE . INSPECT the IWE clutch ring, hub engagement teeth, and the wheel bearing for wear or damage. INSTALL new components as necessary. REFER to Integrated Wheel End (IWE).

T12 CHECK THE LH IWE OPERATION

- Connect a hand vacuum pump with gauge to the larger port of the LH IWE .
- Observe the LH halfshaft while rotating the LH front tire. Apply between 152 and 203 mm (6 to 8 in) Hg of vacuum and release the vacuum to manually disengage and engage the LH IWE .

Did the LH IWE disengage between 152 and 203 mm (6 to 8 in) Hg and engage when the vacuum is released?

Yes	The IWE system is operating correctly at this time. RECONNECT all IWE system vacuum connections. Test the system for normal operation. GO to Pinpoint Test U
No	REMOVE the LH IWE . INSPECT IWE clutch ring, hub engagement teeth, and the wheel bearing for wear or damage. INSTALL new components as necessary.. REFER to Integrated Wheel End (IWE).

Pinpoint Test U: Four-Wheel Drive (4WD) System Functional Test

Refer to Wiring Diagrams Cell [34](#) , Four-Wheel Drive Systems for schematic and connector information.

Locator views for the vacuum lines and vacuum reservoir are located in Description and Operation. Refer to Section 308-07A.

Normal Operation

When operating in 4WD modes, the transfer case clutch is energized and both IWEs are locked. When in 4L mode, the transfer case low range gear set provides an additional 2.64 gear reduction. When operating in 2H mode, the transfer case clutch is de-energized and both IWEs are unlocked.

This pinpoint test is intended to diagnose the following:

- Transfer case and related components
- IWEs and related components
- Wheel/tire assemblies

PINPOINT TEST U : FOUR-WHEEL DRIVE (4WD) SYSTEM FUNCTIONAL TEST

NOTE: 4X4 High and 4X4 Low are not intended for driving on hard/dry surfaces; using these modes while driving on these surfaces may produce noises like intermittent clunking, but will not damage driveline components.

U1 CHECK 2H MODE OPERATION

- Shift the transmission into the NEUTRAL position.
- Switch the MSS to 2H mode.

Are there any 4WD mode indicators present in the IPC message center?

Yes	GO to U2 .
No	GO to U4 .

U2 CHECK FOR TCCM DTCS

- Shift the transmission into the PARK position.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self-Test — TCCM .

Are DTCs retrieved?

Yes	REFER to the TCCM DTC Chart in this section for diagnosis.
No	GO to U3 .

U3 CHECK FOR WIND-UP IN 2H MODE

- Drive the vehicle on a dry, hard surface while executing turns.

Is wind-up present in turns?

Yes	GO to Pinpoint Test P .
No	GO to Pinpoint Test V .

U4 CHECK AUTO MODE OPERATION

- Switch the MSS to 4A mode.

Is the 4X4 auto (4A) mode indicator on in the IPC message center?

Yes	GO to U7 .
No	GO to U5 .

U5 CHECK FOR TCCM DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self-Test — TCCM .

Are DTCs present?

Yes	REFER to the TCCM DTC Chart in this section for diagnosis.
No	GO to U6 .

U6 CHECK FOR WIND-UP IN 4A MODE

- Drive the vehicle on a dry, hard surface, executing turns while applying the throttle.

Is wind-up present in turns?

Yes	GO to U7 .
No	GO to U8 .

U7 CHECK THE TCCM 4WD PWM CLUTCH STATUS (4WD_CLTCH_OUT) PID

- Enter the following diagnostic mode on the scan tool: DataLogger — TCCM .
- Monitor the 4WD_CLTCH_OUT PID.

Does the PID indicate 2 to 10% clutch application at closed throttle?

Yes	GO to U8 .
No	GO to U9 .

U8 CHECK FOR WIND-UP IN 4A MODE WITH 100% CLUTCH USING THE TCCM 4WD PWM CLUTCH STATUS (4WD_CLTCH_OUT) ACTIVE COMMAND

- Enter the following diagnostic mode on the scan tool: DataLogger — TCCM .
- Command the 4WD_CLTCH_OUT to 100%.
- Drive the vehicle on a dry, hard surface, executing turns.

Is wind-up present in turns?

Yes	GO to U9 .
No	CHECK the transfer case. REFER to Section 308-07B.

U9 CHECK THE TCCM AVERAGE FRONT WHEEL SPEED (FRT_WHL_SPD) AND AVERAGE REAR WHEEL SPEED (REAR_WHL_SPD) PIDS

- Enter the following diagnostic mode on the scan tool: DataLogger — **TCCM** .
- Monitor the FRT_WHL_SPD and REAR_WHL_SPD PIDs.
- Drive the vehicle 30 km/h (18 mph).
- Record and compare the FRT_WHL_SPD and REAR_WHL_SPD PIDs.

Do the FRT_WHL_SPD and REAR_WHL_SPD PID values match within 1.5 km/h (1 mph)?

Yes	GO to U10 .
No	VERIFY all 4 tires are the same size and brand. INSTALL new tires as necessary. If the tires are OK, CHECK the ABS. REFER to Section 206-09 and VERIFY that the 4 road tire sizes match. TEST the system for normal operation.

U10 CHECK THE TCCM ABSOLUTE THROTTLE POSITION (TP_TCCM) PID

- Enter the following diagnostic mode on the scan tool: DataLogger — **TCCM** .
 - Ignition ON.
 - Monitor the TP_TCCM PID.
 - **NOTE:** *The TP_TCCM PID is **APP**, as communicated by the PCM via **HS-CAN** .*
- Slowly move throttle from closed throttle to **WOT** and observe the TP_TCCM PID.

Does the TP_TCCM PID value vary between 0% at closed throttle and 100% at **WOT ?**

Yes	GO to U11 .
No	CHECK the APP sensor and/or the PCM. REFER to the Powertrain Control/Emissions Diagnosis (PC/ED) manual.

U11 CHECK 4H MODE OPERATION

- Switch the **MSS** to 4H mode.

Does the 4X4 (4H) indicator illuminate in the **IPC message center correctly?**

Yes	GO to U14 .
No	GO to U12 .

U12 CHECK FOR TCCM DTCS

- Shift the transmission into the PARK position.
- Enter the following diagnostic mode on the scan tool: Self-Test — TCCM .

Are DTCs retrieved?

Yes	REFER to the TCCM DTC Chart in this section for diagnosis.
No	GO to U13 .

U13 CHECK FOR WIND-UP IN 4H MODE

- Drive the vehicle on a dry, hard surface while executing turns.

Is wind-up present in turns?

Yes	GO to Pinpoint Test Q .
No	GO to Pinpoint Test O .

U14 CHECK FOR STRAIGHTLINE WIND-UP IN 4H MODE

- Drive the vehicle on a dry, hard surface.

Is straightline wind-up present?

Yes	GO to Pinpoint Test S .
No	GO to U15 .

U15 CHECK WIND-UP IN TURNS IN 4H MODE

- Drive the vehicle in 4H mode while turning.

Is wind-up present in turns?

Yes	GO to U16 .
No	GO to Pinpoint Test O .

U16 CHECK 4L MODE OPERATION

- Bring the vehicle to a stop with the transmission in NEUTRAL, switch the MSS to 4L mode.

Does the 4WD LOW (4L) indicator in the IPC message center illuminate correctly?

Yes	GO to U19 .
No	GO to U17 .

U17 CHECK FOR TCCM DTCS

- Shift the transmission into the PARK position.
- Enter the following diagnostic mode on the scan tool: Self-Test — **TCCM**.

Are DTCs retrieved?

Yes	REFER to the TCCM DTC Chart in this section for diagnosis.
No	GO to U18 .

U18 CHECK FOR WIND-UP IN 4L MODE

- Drive the vehicle on a dry, hard surface while executing turns.

Is wind-up present in turns?

Yes	GO to Pinpoint Test V .
No	GO to Pinpoint Test P .

U19 CHECK FOR WIND-UP IN TURNS IN 4L MODE

- Drive the vehicle on a dry, hard surface while executing turns.

Is wind-up present and 4L mode functioning?

Yes	GO to U20 .
No	CHECK for DTCs. REFER to the Inspection and Verification procedure in this section. If no DTCs are present, GO to Pinpoint Test P .

U20 CHECK 4L MODE TO 2H MODE SHIFT

- With the vehicle at a stop and transmission in NEUTRAL, switch the **MSS** from 4L mode to 2H mode.

Are there any **4WD** mode indicators present in the message center?

Yes	GO to U21 .
No	GO to U23 .

U21 CHECK FOR TCCM DTCS

- Shift the transmission into the PARK position.
- Release the brake pedal.
- Enter the following diagnostic mode on the scan tool: Self-Test — **TCCM**.

Are DTCs retrieved?

Yes	REFER to the TCCM DTC Chart in this section for diagnosis.
No	GO to U22 .

U22 CHECK FOR WIND-UP IN 2H MODE

- Drive the vehicle on a dry, hard surface while executing turns.

Is wind-up present in turns?

Yes	GO to Pinpoint Test P.
No	GO to Pinpoint Test V.

U23 CHECK FOR TRANSFER CASE DISENGAGEMENT

- With the vehicle in NEUTRAL, position it on a hoist. Refer to Section 100-02.
- Rotate the rear driveshaft while observing the front driveshaft.

Does the front driveshaft rotate?

Yes	REFER to Section 308-07B.
No	GO to U24 .

U24 CHECK FOR LH AND RH DISENGAGEMENT

NOTE: The engine must be at idle during the following steps to supply vacuum for ~~IWE~~ engagement/disengagement.

- Start the engine and allow to idle.
- Rotate the LF tire one revolution forward and one revolution backward while observing the LF halfshaft and U-joint.
- Rotate the RF tire one revolution forward and one revolution backward while observing the RF halfshaft and U-joint.

Does the LH or RH front halfshaft rotate?

Yes	GO to Pinpoint Test N.
No	GO to U25 .

U25 CHECK FOR VACUUM LEAKS

- Ignition OFF.
- Disconnect both vacuum lines from the ~~IWE~~ solenoid.

- Using a hand vacuum pump, apply 508 mm (20 in) Hg of vacuum to the **IWE** apply vacuum circuit.
- Observe the vacuum reading.

Did the vacuum drop more than 25 mm (1 in) Hg per minute?

Yes	CHECK the IWE and related components for leaks. GO to Pinpoint Test N.
No	GO to U26 .

U26 CHECK 2H MODE TO 4H MODE SHIFT AT SPEED ABOVE 32 KM/H (20 MPH)

- Switch the **MSS** to 2H mode.
- Drive the vehicle on the road and shift from 2H mode to 4H mode while driving above 32 km/h (20 mph).

Does the transfer case shift to 4H mode satisfactorily?

Yes	The 4WD system is functioning correctly.
No	GO to Pinpoint Test Q.

Pinpoint Test V: The 4X4 Mode Indicators Do Not Operate Correctly/Do Not Operate

Refer to Wiring Diagrams Cell [34](#) , Four-Wheel Drive Systems for schematic and connector information.

Normal Operation

The **4WD** system status is transmitted to the **IPC** from the **TCCM** through the **HS-CAN** to the **4WD** indicator. A steady indicator displays if the vehicle is in 4X4 auto (4A), 4X4 (4H) or 4X4 low (4L). When in **2WD** , the message center will display 4X2 for approximately 4 seconds, then the message center will no longer provide **4WD** system indication until another shift mode is attained. A system fault condition will illuminate the powertrain malfunction indicator. The powertrain malfunction indicator includes other general powertrain system failures. For information about **HS-CAN** , refer to Section 418-00.

This pinpoint test is intended to diagnose the following:

- **IPC**
- **4WD** system fault
- **HS-CAN** circuits
- **TCCM**

PINPOINT TEST V : THE 4X4 MODE INDICATORS DO NOT OPERATE CORRECTLY/DO NOT OPERATE

V1 CHECK FOR TCCM DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self-Test — **TCCM** .

Are DTCs present?

Yes	REFER to the TCCM DTC Chart in this section for diagnosis.
No	GO to V2 .

V2 CHECK FOR IPC DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self-Test — **IPC** .

Are DTCs present?

Yes	REFER to Section 413-01.
No	GO to V3 .

V3 CHECK 4WD SYSTEM OPERATION

- Test drive the vehicle and operate the **4WD** system.

Does the **4WD** system operate correctly?

Yes	REFER to Section 413-01.
No	GO to Symptom Chart — Four Wheel Drive (4WD) in this section.

Pinpoint Test W: Vehicle Binds in a Turn/Resists Turning/Pulsates in a Straightline in **4WD** Mode

Normal Operation

When the vehicle is operating in **4WD** modes, all 4 wheels receive the same amount of torque and rotate at the same speed. The amount of traction the vehicle can achieve depends on the amount of traction each tire can establish with the road surface. In order for the **4WD** system to function correctly, each tire must be in good condition and the front and rear axle ratios must match.

This pinpoint test is intended to diagnose the following:

- Unmatched tire sizes
- Unequal amounts of tire wear
- Unequal tire inflation pressures
- Throttle position failure
- Transfer case
- **TCCM**

PINPOINT TEST W : VEHICLE BINDS IN A TURN/RESISTS TURNING/PULSATES IN A STRAIGHTLINE IN 4WD MODE

NOTE: *4X4 is not intended for driving on hard/dry surfaces; using 4X4 mode while driving on these surfaces may produce noises like intermittent clunking, but will not damage driveline components.*

W1 VERIFY THE CONCERN

- Test drive the vehicle on a straight stretch of road in 4A mode.

- Drive the vehicle through turns and parking lot maneuvers.

Does the vehicle bind, pulsate or shudder?

Yes	If the problem occurs in 4A mode, GO to W2 ; If 4H mode or 4L mode, binding is a normal condition.
No	GO to Pinpoint Test S.

W2 CHECK THE 4H AND 4L INDICATORS

- Review the test drive notes.

Did the 4X4 (4H) or 4X4 LOW (4L) or powertrain malfunction indicator (wrench light) illuminate in the IPC message center?

Yes	If the powertrain malfunction indicator (wrench light) illuminated, GO to W4 . If the 4H mode or 4L mode indicators illuminated, GO to W3 .
No	GO to W5 .

W3 VERIFY THE FAULT

- Review the test drive notes.

Was 4H mode or 4L mode selected when the 4H or 4L indicators were illuminated in the IPC message center?

Yes	The system is functioning correctly.
No	The fault may have been caused by an intermittent, uncommanded or autolock shift. TEST the system for normal operation. GO to Pinpoint Test S.

W4 CHECK FOR TCCM DTCS

- Enter the following diagnostic mode on the scan tool: Self-Test — TCCM .

Are DTCs retrieved?

Yes	REFER to the TCCM DTC Chart for diagnosis.
No	GO to W5 .

W5 VERIFY THE CONCERN WITH TRANSFER CASE CONNECTOR DISCONNECTED

- Disconnect: Transfer Case .
- Test drive the vehicle on a straight stretch of road and in turns, and shift the transfer case through each mode. Record any fault conditions.

Is the concern present?

Yes	INSPECT the driveline components for damage or wear. REFER to Inspection and Verification in this section. TEST the system for normal operation.
No	GO to W6 .

W6 CHECK THE TCCM 4WD PWM CLUTCH STATUS (4WD_CLTCH_OUT) PID

- Connect: Transfer Case .
- Using the scan tool, clear the DTCs.
- Enter the following diagnostic mode on the scan tool: DataLogger — TCCM .
- Monitor the 4WD_CLTCH_OUT PID at closed throttle, K.O.E.O. .
- Select 4H mode.

Does the PID indicate 2 to 10% clutch application at closed throttle?

Yes	GO to W7 .
No	CHECK the transfer case. REFER to Section 308-07B. TEST the system for normal operation.

W7 CHECK THE ABS MODULE WHEEL SPEED SENSOR PIDS

- Enter the following diagnostic mode on the scan tool: DataLogger — ABS Module .
- While driving the vehicle at 30 km/h (18 mph), monitor the following wheel speed sensor PIDs:
 - Left Front Wheel Speed Sensor (LF_WSPD)
 - Left Rear Wheel Speed Sensor (LR_WSPD)
 - Right Front Wheel Speed Sensor (RF_WSPD)
 - Right Rear Wheel Speed Sensor (RR_WSPD)
- Compare the speedometer reading to the wheel speed sensor PIDs.

Do the wheel speed sensor PIDs and speedometer speeds match within 1.5 km/h (1 mph)?

Yes	GO to W8 .
No	VERIFY that the 4 road wheels are the same size and have similar tread depth. CHECK the ABS for faults. REFER to Section 206-09. TEST the system for normal operation.

W8 CHECK THE TCCM ABSOLUTE THROTTLE POSITION (TP_TCCM) PID

- Enter the following diagnostic mode on the scan tool: DataLogger — TCCM .
- **NOTE:** *The TP_TCCM PID is APP, as communicated by the PCM via HS-CAN .*
Slowly move accelerator pedal from closed throttle to and observe the TP_TCCM PID.

Does the TP_TCCM PID value vary between 0% at closed throttle and 100% at W.O.T ?

Yes	GO to W9 .
No	DIAGNOSE the PCM. REFER to the Powertrain Control/Emissions Diagnosis (PC/ED) manual. TEST the system for normal operation.

W9 CHECK THE TRANSFER CASE FRONT DRIVESHAFT OUTPUT

- With the vehicle in NEUTRAL, position it on a hoist. Refer to Section 100-02.
- Rotate the front driveshaft by hand.

Does the front driveshaft rotate freely (does it require less than 45 Nm [33 lb-ft] to rotate)?

Yes	GO to W10 .
No	CHECK the transfer case. REFER to Section 308-07B. TEST the system for normal operation.

W10 CHECK FOR CORRECT TCCM OPERATION

- Ignition OFF.
- Disconnect and inspect all ~~TCCM~~ connectors.
- Repair:
 - corrosion (install new connector or terminals - clean module pins)
 - damaged or bent pins - install new terminals/pins
 - pushed-out pins - install new pins as necessary
- Connect all ~~TCCM~~ connectors. Make sure they are seated correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new TCCM . REFER to Transfer Case Control Module (TCCM) in this section. TEST the system for normal operation.
No	The system is operating correctly at this time. Concern may have been caused by a loose or corroded connector. ADDRESS the root cause of the connector or pin issues.

Pinpoint Test X: The Vehicle Does Not Shift Between 2H (2WD) and 4X4 Modes Correctly

Refer to Wiring Diagrams Cell [34](#) , Four-Wheel Drive Systems for schematic and connector information.

Normal Operation

The ~~MSS~~ communicates the operator's choice to the ~~TCCM~~ . The ~~TCCM~~ then controls the transfer case synchronization clutch, transfer case motor and ~~IWE~~s as necessary. If the vehicle is not responding as intended, systematically check the necessary inputs and outputs of the ~~TCCM~~ , components of the transfer case, ~~IWE~~ components and halfshafts. Check all circuits for opens and shorts to power or ground.

DTC Description	Fault Trigger Condition
P1812:00 — 4-Wheel Drive Mode Select Circuit Failure: No Sub Type Information	Fault sets when TCCM C2371A pin 16 senses an open circuit. 4WD system mode cannot be changed.
P1815:00 — 4-Wheel Drive Mode Select Short To Ground: No Sub Type Information	Fault sets when TCCM C2371A pin 16 detects a short to ground. 4WD system mode cannot be changed.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- ~~MSS~~
- Transfer case
- ~~TCCM~~

PINPOINT TEST X : THE VEHICLE DOES NOT SHIFT BETWEEN 2H (2WD) AND 4X4 MODES CORRECTLY

WARNING: When directed to drive the vehicle as part of this test, drive the vehicle on a hard surface in an area without traffic to prevent a crash. Failure to follow these instructions may result in personal injury.

X1 CHECK FOR TCCM DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self-Test — ~~TCCM~~ .

Are any DTCs present?

Yes	For DTC P1812:00, GO to X5 . For DTC P1815:00, GO to X7 . For all other DTCs, REFER to the DTC Chart in this section.
No	GO to X2 .

X2 CHECK THE TCCM IGNITION POSITION (KEY_POS_4X4) AND IGNITION STATUS (KEY_STATUS_4X4) PIDS

- Enter the following diagnostic mode on the scan tool: DataLogger — ~~TCCM~~ .
- Monitor the KEY_POS_4X4 and KEY_STATUS_4X4 PIDs while cycling through each position.

Do the ignition PIDs match the ignition switch positions?

Yes	GO to X3 .
No	If the KEY_STATUS_4X4 PID did not match, CHECK BJB fuse 54 (5A). If the fuse is OK, REPAIR circuit CBB54 (VT/OG). If the fuse is not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. After the fault is repaired, CLEAR the DTCs. TEST the system for normal operation. If the KEY_POS_4X4 PID did not match, REFER to Section 211-05 to diagnose the ignition switch. After the fault is repaired, CLEAR the DTCs. TEST the system for normal operation.

X3 CHECK THE TCCM MODULE SUPPLY VOLTAGE (VBAT_4X4) PID

- Start the engine.
- Enter the following diagnostic mode on the scan tool: DataLogger — TCCM .
- Monitor the VBAT_4X4 PID.

Is the VBAT_4X4 PID greater than 12 volts?

Yes	GO to X4 .
No	REFER to Section 414-00 to DIAGNOSE the low battery voltage and/or charging system concern. CLEAR the DTCs. TEST the system for normal operation.

X4 CHECK THE MSS POSITION MONITORING THE TCCM 4WD SWITCH STATUS (4X4_SW) PID

- Enter the following diagnostic mode on the scan tool: DataLogger — TCCM .
- Monitor the 4X4_SW PID while switching to each MSS position.

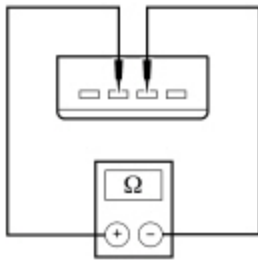
Does the PID match the MSS position?

Yes	GO to X10 .
No	GO to X5 .

X5 CHECK THE MSS RESISTANCE VALUES

- Ignition OFF.
- Disconnect: MSS .
- Measure the resistance between MSS C284 pins 2 and 3, component side, while rotating the MSS through all positions. Refer to the following chart.

MSS Position	Resistance (±1 %)
2H	619 ohms
4A	270 ohms
4H	130 ohms
4L	62 ohms



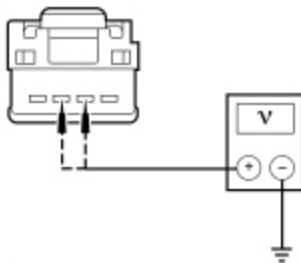
• N0063005

Are the resistance values within specification?

Yes	GO to X6 .
No	INSTALL a new MSS . REFER to Mode Select Switch (MSS) in this section. CLEAR the DTCs. TEST the system for normal operation.

X6 CHECK THE MSS CONTROL AND RETURN CIRCUITS FOR SHORT TO VOLTAGE

- Disconnect: TCCM [C2371A](#) .
- Ignition ON.
- Measure the voltage between MSS [C284](#) Pin 3, circuit RCF13 (WH/BN), harness side and ground; and between MSS [C284](#) Pin 2, circuit CCF13 (WH/BU), harness side and ground.



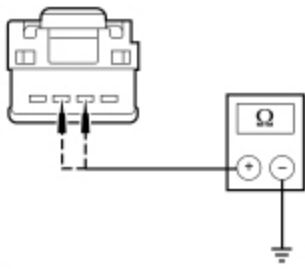
• A0092316

Is any voltage present?

Yes	REPAIR the affected circuit(s). CLEAR the DTCs. TEST the system for normal operation.
No	GO to X7 .

X7 CHECK THE MSS CONTROL AND RETURN CIRCUITS FOR SHORT TO GROUND

- Ignition OFF.
- Disconnect: TCCM [C2371A](#) .
- Disconnect: MSS .
- Measure the resistance between MSS [C284](#) Pin 3, circuit RCF13 (WH/BN), harness side and ground; and between MSS [C284](#) Pin 2, circuit CCF13 (WH/BU), harness side and ground.



• A0092317

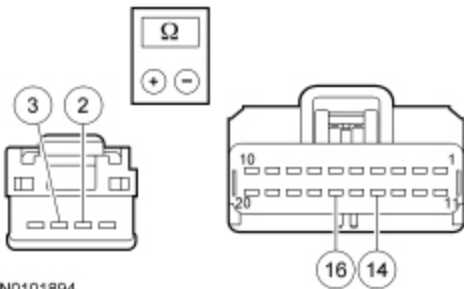
Is the resistance greater than 10,000 ohms?

Yes	GO to X8 .
No	REPAIR the affected circuit(s). CLEAR the DTCs. TEST the system for normal operation.

X8 CHECK THE MSS CONTROL AND RETURN CIRCUITS FOR AN OPEN

- Disconnect: ~~TCCM~~ [C2371A](#) .
- Measure the resistance between MSS [C284](#), and ~~TCCM~~ [C2371A](#), harness side. Refer to the following chart.

MSS	Circuit	TCCM
C284 Pin 3	RCF13 (WH/BN)	C2371A Pin 14
C284 Pin 2	CCF13 (WH/BU)	C2371A Pin 16



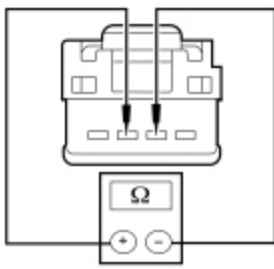
• N0101894

Are the resistances less than 3 ohms?

Yes	GO to X9 .
No	REPAIR the affected circuit(s). CLEAR the DTCs. TEST the system for normal operation.

X9 CHECK THE MSS CONTROL AND RETURN CIRCUITS FOR A SHORT TOGETHER

- Ignition OFF.
- Measure the resistance between MSS [C284](#) Pin 3, circuit RCF13 (WH/BN), harness side and MSS [C284](#) Pin 2, circuit CCF13 (WH/BU), harness side.



- N013253B

Is the resistance greater than 10,000 ohms?

Yes	GO to X10 .
No	REPAIR the affected circuit(s). CLEAR the DTCs.

X10 CHECK FOR CORRECT TCCM OPERATION

- Ignition OFF.
- Disconnect and inspect all **TCCM** connectors.
- Repair:
 - corrosion (install new connector or terminals - clean module pins)
 - damaged or bent pins - install new terminals/pins
 - pushed-out pins - install new pins as necessary
- Connect all **TCCM** connectors. Make sure they are seated correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new TCCM . REFER to Transfer Case Control Module (TCCM) in this section. TEST the system for normal operation.
No	The system is operating correctly at this time. Concern may have been caused by a loose or corroded connector. ADDRESS the root cause of the connector or pin issues.